Heriot-Watt University

**MASTERS THESIS**

# Self-Supervised Learning with Equivariant Imaging (EI) for Dynamic MRI Reconstruction Inverse Problem

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**Abstract**

Current state-of-the-art Magnetic Resonance Imaging (MRI) reconstruction methods rely on supervised learning models that necessitate ground truth data. However, it is often expensive to obtain such data due to the long acquisition time. In addition, a fully-sampled MRI acquisition requires the patient to stay immobile during the whole process which is almost unfeasible and potentially yields motion artifacts that decrease the reconstruction quality. To reduce acquisition time, it is common practice to capture less information by using accelerated scans. These accelerated scans are paired with ground truth scans in order to learn a reconstruction model. However, we might not always be in the possession of a set of ground truth scans as it is expensive to acquire!

Recently, the introduction of the cutting edge Equivariant Imaging (EI) and its variant Robust EI (REI) proposed a method to overcome this issue by exploiting some prior information contained in a measurement's transformations. The approach has been applied to different image reconstruction problems such as Computed Tomography, Magnetic Resonance Imaging, inpainting..., but always in 2D. Current experiments demonstrated that EI results are comparable to common paired-learning algorithms but with measurements only. This research explores the applicability of EI method to reconstruct Dynamic MRI sequences from subsampled k-space raw data measurements only, through a self-supervised learning (SSL) algorithm exploiting diverse transformations.

We introduce a new transformation specific to dynamic problems with the temporal symmetry over the time axis and propose a smart method to avoid saturating the GPU memory even in full resolution. We propose different models based on a Plug-and-Play (PnP) prior with Denoising Convolutional Neural Network (DnCNN) whose parameters are learned through an unrolling framework.

The models are evaluated on pre-processed data from the QTIM dataset we degrade with Gaussian random cartesian masks to simulate MRI acquisition with different noise levels. We use the Peak signal-to-noise ratio (PSNR) metric over the sequences to measures the accuracy of the models.
Regarding the results obtained, we compete with fully supervised networks models using the ground truth signal during the learning process. At low noise level it is interesting to use the vanilla EI model while the robust version shows outstanding results at high noise level. Obtaining these good reconstructions is a first step towards the reconstruction of functional MRI (fMRI) sequences which is inherently dynamic.

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# Chapter 5

# Learning without Ground-Truth

## 5.1 Formulation of the Problem

Since the beginning of this report, when we explored MRI reconstruction methods, it was only through supervised-learning techniques that are based on the existence of numerous collected ground-truth images. However, this is an extremely hard assumption which is very criticizable. In inverse problems, we are absolutely not supposed to already have ground-truth images that can be used to have pairs of measurements and outputs. One even could consider that "ground-truth" data are fundamentally inaccessible in inverse problems since our aim is to find solutions close to the ground trutH Frequently, whether in astronomy or medical imaging, we only get some degraded measurements. Especially, in MRI, where the forward model $A$ has a non-trivial null-space, it's theoretically impossible to learn the signal using only compressed samples, and nobody has even seen a "true" MRI image. Then, we need to have algorithms that can work without any ground-truth data. This can be achieved through self-supervised learning algorithms which rely only on $y_i$ measurements. This is fundamentally way more challenging as there is much more information in $x_i$ "real" outputs than $y_i$ measurements only, because of the challenge to see inside $A$ null-space. Thus, it seems impossible to learn from compressed data without imposing strong assumptions. We have to find a way to reinject information in the learning process, without nothing more than measurements.
Our learning task consists in learning a reconstruction function (inverse mapping) $h_\theta(y) = x$ from a collection of measurements $y_i$ only. The learnable solver $h_\theta$ should thus to be consistent with measurements. For all $y_i$,

$$Ah_\theta(y_i) = y_i \tag{5.1}$$

Let's recall that the linear imaging physics model $A$ (from $\mathbb{R}^n$ to $\mathbb{R}^m$ with $m < n$) has a nontrivial null-space $\mathcal{N}_A \subseteq \mathbb{R}^n$ whose dimension $dim(\mathcal{N}_A) \geq n - m$ (see 3.5). By definition of the null-space, we cannot expect to learn any information outside the range of operator $A$ $\mathcal{R}_A$ from measurements only. Indeed, from proposition 1 of [4] any mapping $A^+ + v$ where $v$ has images $\in \mathcal{N}_A$ will also satisfy the consistency requirement, leading to an infinite number of solutions (including pseudo inverse $A^+$). The challenge amounts thus to get information about the null-space without ground truth data.

Since the measurement consistency is here not sufficient to efficiently reconstruct the original set of signals $\mathcal{X}$, as lot of functions $h_\theta$ can satisfy the equation (even with an infinite number of samples $y_i$), we have to consider other assumptions such as the invariances properties of $\mathcal{X}$ to retrieve information about $\mathcal{N}_A$. This is based on the assumption that the original signals are invariant to certain transformations, as flips or/and rotations. Especially, in medical imagery, the human body is assumed to be roughly symmetrical at tissue scale and the purpose is to exploit this mild prior to gain information. Therefore, $\mathcal{X}$ is assumed to be invariant to a specific group of transformations (see section 6.2) $\mathcal{G} = \{g_1, g_2, \ldots, g_{|\mathcal{G}|}\}$ which are unitary matrices $T_g \in \mathbb{R}^{n \times n}$ such that, $\forall x \in \mathcal{X}$ and $\forall g \in \mathcal{G}$ we have [4]:

$$T_g x \in \mathcal{X} \tag{5.2}$$

This directly leads to the composition $h \circ A$ being equivariant to $T_g$ transformations for same conditions:

$$h\left(AT_g x\right) = T_g h(Ax) \tag{5.3}$$

Let's insist on the required composition $h \circ A$ to be equivariant, and not of $h$ to be invariant or equivariant. As a reminder, invariance refers to a property of a function where the output remains unchanged despite variations or transformations in the input, and equivariance refers to a property of a function where the output undergoes a corresponding transformation when the input is transformed. Thus, invariance describes $f\left(T\left(x\right)\right) = f\left(x\right)$ while equivariance describes $f\left(T\left(x\right)\right) = T^{'}\left(f\left(x\right)\right)$ where $T'$ represents a corresponding transformation on the output space. We will see later the importance of having a denoiser $\mathcal{D}$: $\mathcal{X} \to \mathcal{X}$ satisfying group equivariance property when the dataset of images has a group invariance property.

## 5.2 Equivariant Imaging (general)

### 5.2.1 Principle of Equivariant Imaging

An attempt to overcome the challenge of learning without ground-truth is proposed by Dr. Chen and his colleagues through the cutting-edge Self-Supervised Learning with Equivariant Imaging (SSLEI) technique [4]. Let's be clear on one aspect, equivariance (therefore EI) can be imposed in image reconstructions problems by two manners. We distinguish the equivariance by design, meaning the architecture of the model considers equivariance, from the equivariance by learning, meaning we enforce it through the training process. This second approach also regroups two types of learning: with ground truth and therefore using data augmentation (DA) which is not what we want, or by self-supervised learning (SLL). We insist on equivariance by learning in our research as it can be applied to a wide variety of machine learning models, but especially, in regard to what we explained before, on SSLEI.
Equivariance by design can be achieved thanks to model-based models incorporating priors invariant to certain types of transformations, such as TV which is invariant to translations, orthogonal rotations and flips, or wavelets models which are invariant to translations and scale. Equivariance by design also includes the use of equivariant neural networks, i.e. denoisers incorporating equivariant property (unlike DnCNN which is not inherently equivariant).
Equivariance by learning method is based on the assumption that, even if we don't directly get information about the null-space of $A$, valuable insights can still be extracted with transformations $T_g$ on the data thanks to the $\mathcal{X}$ invariant set consistency. Therefore, the aim at using equivariance in the loss of an optimisation problem in SSLEI is to leverage the inherent symmetries and rotations of images to infer missing information and enhance reconstruction quality. Authors in [4] showed that by exploiting the inherent invariances to certain transformations in the imaging process (shifts, rotations...), equivariant imaging techniques can effectively recover structural details and features that may be obscured or distorted in the acquired data.

### 5.2.2 Theoretical Analysis: Seeing the Null-Space

In the section 2.2, we showed that self-supervised learning of a MRI solver $h_\theta$ is impossible without using strong assumptions on the set of possible solutions. Therefore, working in a framework without invariance, if we don't have access to the ground-truth measurements, there is no information we can extract to learn the null-space. However, in the context of EI, each measurement can be seen as a new observation with a new measurement operator $A_g = AT_g$:

$$y = Ax = AT_g T_g^T x \tag{5.4}$$

Using $\tilde{x} = T_g^T x$, also a signal in $\mathcal{X}$, yields $y = A_g \tilde{x}$ and the invariance property (see section 5.1) allows us to see in the range of the operators $A_g$ and not only in $\mathcal{R}_A$. This can be understood as the invariance rotates $\mathcal{R}_A$ under the action of $\mathcal{G} = \left\{g_1, g_2, \ldots, g_{|\mathcal{G}|}\right\}$ such as:

$$\mathcal{R}_{A_g} = range\left(T_g^T A^T\right) = T_g^T \mathcal{R}_A \tag{5.5}$$

Meaning we now see in a bigger range space $\cup_g \mathcal{R}_{A_g} \supseteq \mathcal{R}_A$ . Ideally, we would like to reduce the null-space to a trivial one. Therefore, we can deduce the necessary condition to recover $\mathcal{X}$ from compressed measurements, which is that the matrix:

$$M = \begin{bmatrix} AT_1 \\ \vdots \\ AT_{|\mathcal{G}|} \end{bmatrix} \in \mathbb{R}^{|\mathcal{G}|m \times n} \tag{5.6}$$

is of rank $n$. This yields the lower bound on how large $\mathcal{G}$ has to be with $m\,|\mathcal{G}| \geq n$. We need to be vigilant as not any combination of forward operator and group action verify this condition (see section 6.2). A necessary condition is that $\mathcal{R}_A$ can't be invariant to all $g \in \mathcal{G}$ (which is required to learn beyond $\mathcal{R}_A$), therefore there is one $g$ that satisfies: $\mathcal{R}_{A_g} \neq \mathcal{R}_A$.

## 5.3 Equivariant Imaging Losses

We can now incorporate these properties in a model loss function. First, let's consider an unsupervised loss that updates the parameters $\theta$ of neural network:

$$\sum_{i=1}^{N} \|Ah_\theta(y_i) - y_i\|_2^2 \tag{5.7}$$

In [4], authors propose the vanilla equivariant imaging loss, but developed different variations in [6], with the $\text{EI}_1$, $\text{EI}_2$, and REI losses.

### 5.3.1 Vanilla EI Loss

This naïve loss doesn't incorporate any information regarding equivariance of images or their reconstruction in $\mathcal{N}_A$. However, incorporating the mild prior from invariances to actions from the signals, this weak assumption can be sufficient for fully unsupervised learning. The group actions rotating the null-space of $A$ such that $\mathcal{N}_{A_g} = T_g^T \mathcal{N}_A$ can potentially expose parts of the null-space. The loss being composed of the data consistency part and the term to impose equivariance, we try to minimize:

$$\mathcal{L}_{EI}(\theta) = \mathcal{L}_{MC}(\theta) + \lambda \mathcal{L}_{EQ}(\theta) \tag{5.8}$$

Where the measurement consistency is denoted by the error function [6]:

$$\mathcal{L}_{MC}(\theta) = \sum_{i=1}^{N} \frac{1}{m} \|Ah_\theta(y_i) - y_i\|^2 \tag{5.9}$$

And the equivariance error function by:

$$\mathcal{L}_{EQ}(\theta) = \sum_{i=1}^{N} \sum_{g=1}^{|\mathcal{G}|} \frac{1}{|\mathcal{G}|\,n} \|T_g h_\theta(y_i) - h_\theta(AT_g h_\theta(y_i))\|^2 \tag{5.10}$$

Incorporating this into 5.7, the global surrogate loss function we aim to minimize is:

$$\underset{\theta}{\operatorname{argmin}} \sum_{i=1}^{N} \left( \frac{1}{m} \|Ah_\theta(y_i) - y_i\|^2 + \alpha \sum_{g=1}^{|\mathcal{G}|} \frac{1}{|\mathcal{G}|\,n} \|T_g h_\theta(y_i) - h_\theta(AT_g h_\theta(y_i))\|^2 \right) \tag{5.11}$$

Where $\alpha$ is a positive trade-off parameter between the measurement consistency and equivariant loss. This presents the advantage of not using any paired ground-truth data but only the measurements $y_i$ and transformed and reconstructed version of them inside the loss. Experiments showed this loss function gave very good results with low noise and measurements degradation but suffers with high noise.

### 5.3.2 Denoising with SURE and Robust Equivariant Imaging

As the noise increases, the vanilla SSLEI shows poorer results. A possible interpretation is that, while $\ell_{MC}(\theta)$ accounts for the likelihood of noisy measurements $y_i = \overline{y}_i + \epsilon$ where $\overline{y}_i$ stands for an hypothetical perfect measurement, it does not reflect the "additional" noise affecting the solver output $h_\theta(y_i) \neq h_\theta(\overline{y}_i)$. Thus, with the aim to keep satisfying performances with EI despite increasing noise, Robust Equivariant Imaging (REI) [6] has been developed to learn image reconstruction from noisy measurements only. This framework comes from the statement that, in many different real-worlds scenarios, like MRI or CT, observed measurements are corrupted by noise whose distribution is empirically known. Therefore, we can assume a known noise distribution model. From this assumption, if paired models can easily learn a denoiser with a known mapping, Stein's Unbiased Risk Estimator (SURE) proposed a framework using noisy observations only. SURE is an unbiased estimator of the mean squared error of an arbitrary estimator, which, in our case, is $Ah_\theta(y_i)$. SURE method bypasses the need of clean training data with a specific loss (initially for Gaussian noise distribution, then generalized to Poissonian and other...) based on the divergence $\nabla$ of a denoiser. The error function for the SURE loss is initially made of two terms penalizing the error between the noisy observations and denoised ones (bias of the denoiser), and a term penalizing the variance of the denoiser. The SURE loss presents the advantage of being an unbiased estimator, i.e. its expected value is equal to the true value of the parameter. However, this loss requires to compute the divergence of the denoiser function which might not have a simple analytical form. A Monte Carlo method, using randomness to solve problems that might be deterministic in principle, has been proposed to fix this issue by only requiring one additional evaluation of the denoiser.
This SURE method can be exploited in the context of SSLEI but need to be adapted to this specific equivariant framework. The REI model now replaces the loss from [29] with noise-robust terms [6]:

$$\mathcal{L}_{REI}(\theta) = \mathcal{L}_{SURE}(\theta) + \lambda \mathcal{L}_{REQ}(\theta) \tag{5.12}$$

We reproduce in figure 5.1 results of [6] showing the relevance of using the SURE loss with high noise in the case of a knee MRI image. The REI is compared with the ground-truth, a vanilla EI model, and a supervised algorithm, with different simulated Gaussian noise models:

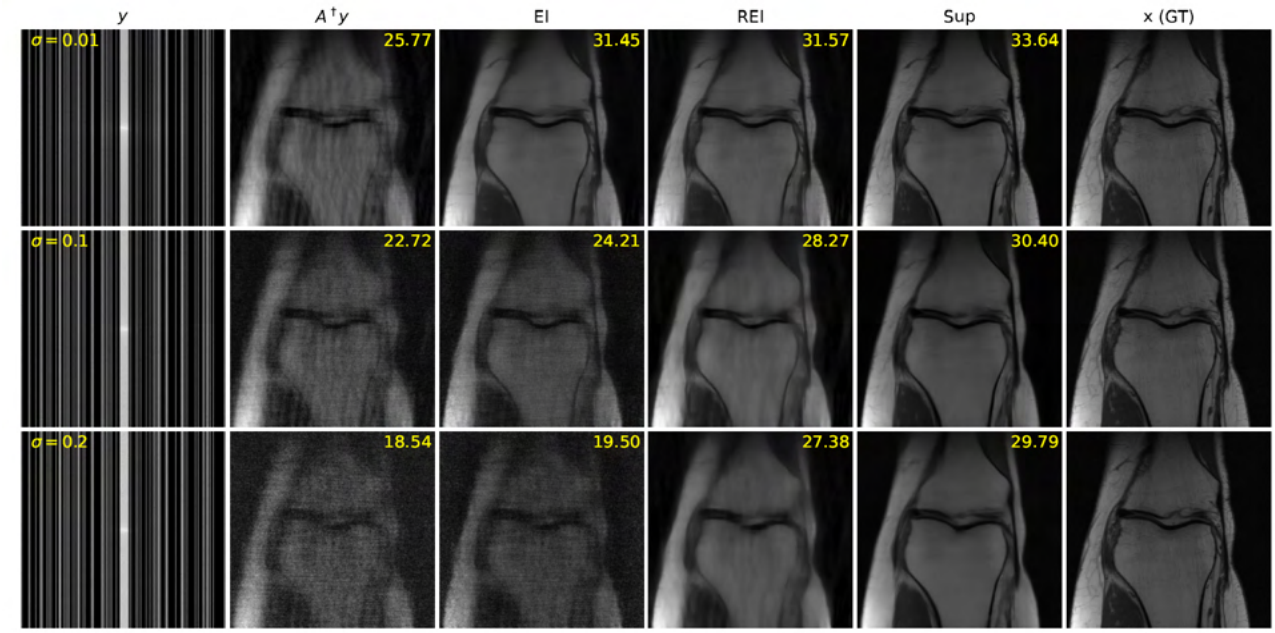



Figure 5.1: MRI image reconstruction on test data with different Gaussian noise and PSNR index [6]

We see that if vanilla EI reconstruction lacks of robustness as the noise is increasing, the REI model achieves results comparable to the supervised model.

In [36] authors propose to enforce robustness of SURE in MRI with an Ensemble-SURE approach when different down-samplings masks are available (but do not make use of EI).

## 5.4 EI Variants

Two variations of EI losses have been proposed, both half-way between vanilla EI and REI.

### 5.4.1 $\text{EI}_1$ Loss

$\text{EI}_1$ conserves the measurement consistency loss $\mathcal{L}_{MC}(\theta)$ but handles the noisy measurements with $\mathcal{L}_{REQ}(\theta)$. The $\text{EI}_1$ Loss complete form is:

$$\mathcal{L}_{EI_1}(\theta) = \mathcal{L}_{MC}(\theta) + \lambda \mathcal{L}_{REQ}(\theta) \tag{5.13}$$

### 5.4.2 $\text{EI}_2$ Loss

The $\text{EI}_2$ conserves the $\mathcal{L}_{EQ}(\theta)$ part but replaces the measurement consistency by the SURE (here Gaussian) loss. This result in:

$$\mathcal{L}_{EI_2}(\theta) = \mathcal{L}_{SURE}(\theta) + \lambda \mathcal{L}_{EQ}(\theta) \tag{5.14}$$

### 5.4.3 Comparison

In [6], experiments with 4x accelerated MRI showed the evolution of the PSNR over the fast MRI knee dataset with increasing noise.

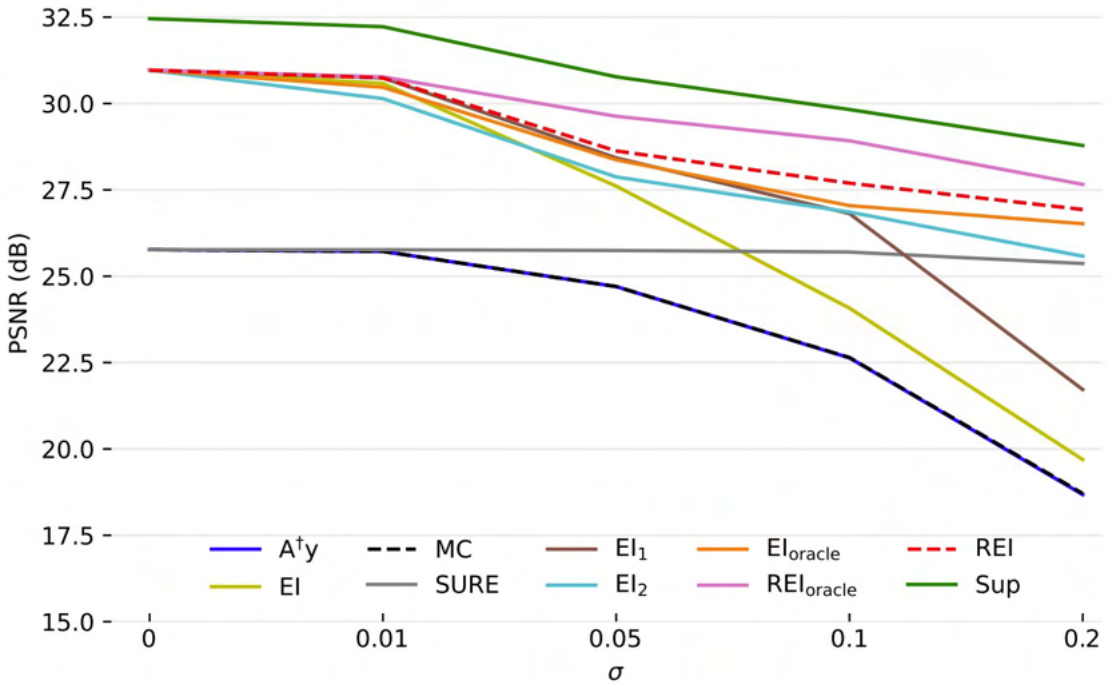



Figure 5.2: EI losses in MRI [6]

We see that both $\text{EI}_1$ and $\text{EI}_2$ overpass vanilla EI, but particularly $\text{EI}_2$ is more stable with higher noise. They still give worse results than REI. It highlights the importance of using the SURE instead of the $\mathcal{L}_{MC}(\theta)$ loss in presence of noise. In absence of noise, REI = $\text{EI}_1$ = $\text{EI}_2$ = EI.

In the rest of the study, by robust models we suggest "REI", and by "EI" we suggest vanilla EI.

# Chapter 6

# Equivariant Imaging for Dynamic MRI

## 6.1 Understanding Dynamic MRI reconstruction

Once we have all the blocks to understand SSLEI, applying it to problems such as image inpainting is relatively intuitive as it directly uses data having the same input and output domain. However, applying the same concept to MRI is more challenging as the measurements are made in the Fourier domain. If 2D-MRI reconstruction has been explored using SSLEI in [1], the introduction of a temporal aspect is new and aims to reconstruct MRI sequences. Limitations linked to this procedure are mentioned in section 7.5. Working on 2D-MRI sequences mean we still have 2 spatial dimensions (defined by Height $H$ and Width $W$), but we use the depth $D$ as a temporal dimension $T$. Therefore, we work on a 2D spatial domain but on a 1D temporal domain.

Similarly to simple 2D MRI, the ground-truth data is altered in the Fourier domain by subsampling some frequencies (Fourier coefficients). The difference is the unfixed aspect of the mask along the temporal axis. In 2D MRI, the same mask is applied over every images of the dataset. In Dynamic MRI, the same mask is applied over every volume of the dataset, but this mask is time-varying over its depth (time). Therefore each slice of the mask volume has a different subsampling, but the same volume mask is applied to each volume.

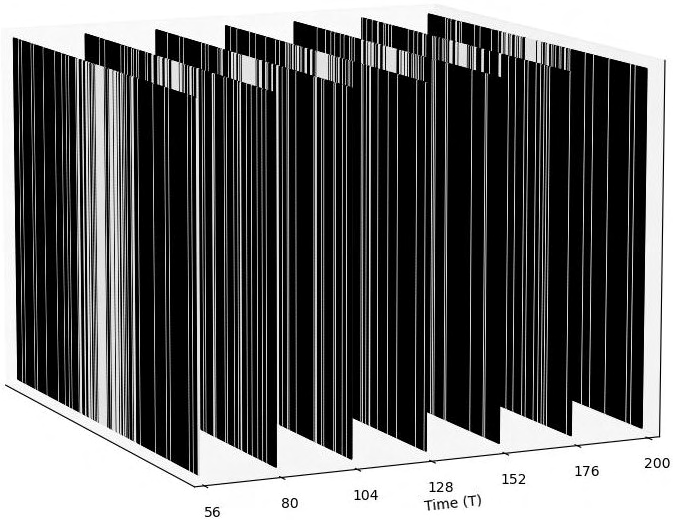



Figure 6.1: Time-Varying MRI Mask

## 6.2 Transformations in Dynamic 2D-MRI Reconstruction

Several transformations have been proposed in EI, and in theory an infinite number exist when combining them. The most common ones we already mentioned are the shifts, flips, rotations, scaling,

and shearing which all belong to the category of linear transformations. This category regroups every type of transformations that can supposedly be applied to recover information about a -supposed- linear transformation operator $A$. We can combine different actions together, such as a flip and a rotation, still resulting in a linear transformation. However, not any transformation can be applied. For instance, shift invariance can't be used to learn from Fourier measurements operators, such as MRI, due to the operator equivariance to translations as $A^T$ is invariant to the shift [4]. Therefore, we need to exclude the translations from our group of transformations.

During the experimentation period, a new set of transformations have been proposed in [5] with Homography. Authors proposed projective transformations with different subgroups. One of them is particularly interesting in MRI, with the "affine" subgroup of transformations. Affine transformations not only include translations and rotations, but also add reflections, skews, and stretches.

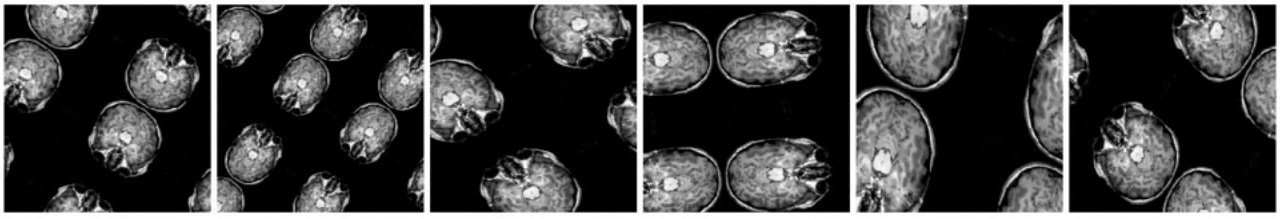
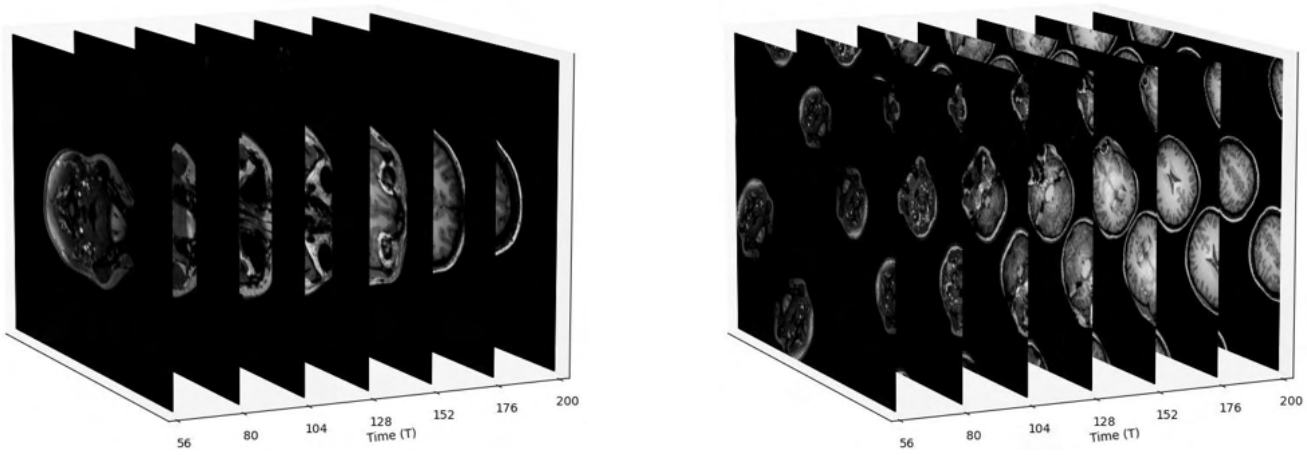


Figure 6.2: Homographic transforms [5]

The human body being intrinsically nearly-symmetrical, rotations and flips only might not help to leverage sufficient information in the EI loss. These transformations mostly conserve the natural symmetry of the body. The addition of cutting-edge projective transformations (pan, tilt, affine...) break the symmetry of a simple rotation and help to leverage information. We can therefore use the Affine homographic group, despite translations that will not affect the reconstruction. As we are now working with 3 dimensions, the spatial transformation must be applied to every slice of the volume. Then, we operate projective transformations over $H$ and $W$ for each slice in $T$.



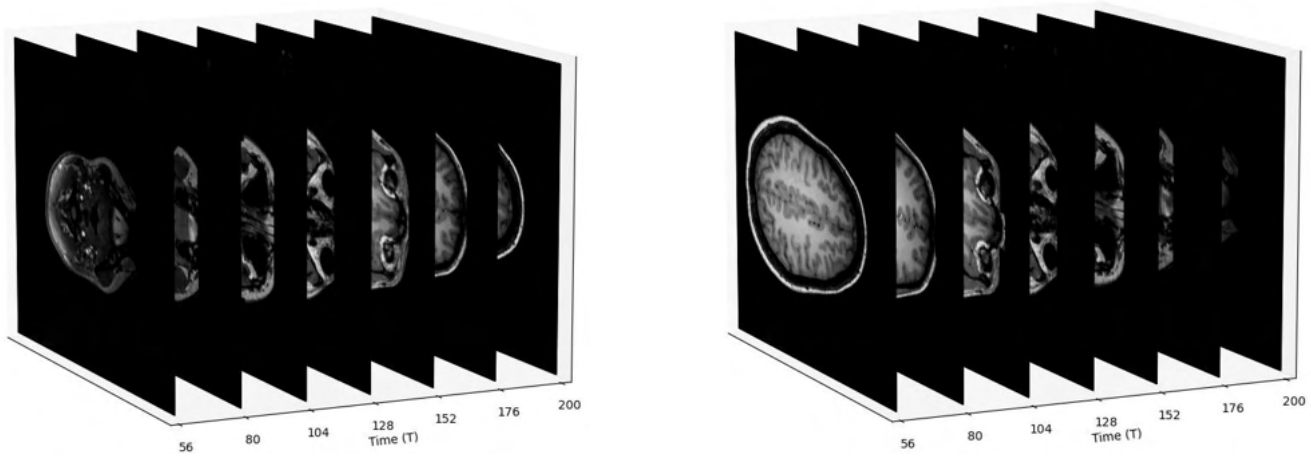

(a) original volume

(b) transformed volume

Figure 6.3: 3D Homographic Transform (no flip)

As for now, the different groups of transformations have only been applied to spatial dimensions in the different research papers. As we innovate by working on dynamic data and not static data, what about the transformations in the time domain? We need to propose transformations satisfying the equivariance of $h \circ A$ in this dimension that make sense.

If in the spatial domain it's intuitive to represent a 2D spatial transformation (e.g. rotation), this is non-sense in the temporal domain which is in one dimension only by definition. How could we rotate over the temporal axis ? Same logic goes with shearing, scaling and other projective transformations that can't be applied to a temporal dimension. What would it mean to scale or shear time? These transformation are not applicable in the time domain, because it's only in one dimension. However, symmetry transformation seems suited for 1D data (no matter in what domain, spatial or temporal).
This is the most evident transformation in the 1D temporal domain, with the flip of our measurement (i.e. sequence of images) alongside the $T$ axis. If we reverse the direction of time, the sequence properties remain the same. The first frame of the sequence becomes the last of the symmetric one, and the last frame of the first sequence becomes the first one of the symmetric one. As the time axis is orthogonal and distinct from the two spatial axis, flipping the time does not impact the spatial transformations and matches the invariance property in temporal 1D.




(a) Original volume

(b) Transformed volume

Figure 6.4: Flipped Volume

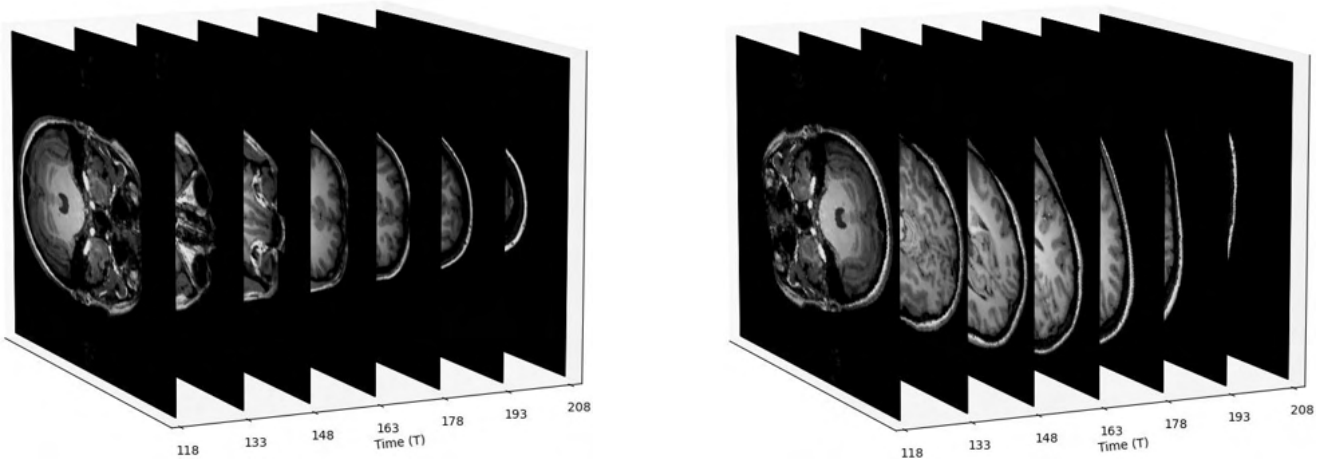
Apart from reversing the time axis, it seems conceivable to propose transformations in 2D+time consisting of a collection of 2D spatial group actions indexed by time preserving spatial invariance property (e.g. 2D rotations whose angle evolves smoothly with time aka helical movement). We are aware that brain anatomical structures can be considered as rigid along the acquisition time. However, in other situations (e.g. cardiac MRI) time deformations of structures seem probable. Even if we do not expect noticeable improvement for dynamic brain images, we suggest now a strategy to build a dynamic collection of spatial group actions.

Let us consider as previously mentioned the group of affine transforms and two elements of this group: $g_0$ and $g_1$ respectively indexed by times $t_0$ and $t_1$ ($t_0 < t_1$). Without loss of generality we admit that $t_0$ indexes the first time slice and $t_1$ the last one. One can then choose (or sample) any $g_0$ in the group (with suitable parameters: rotation, scale, shear) and select $g_1$ "not too far" from $g_0$ by choosing (or sampling) parameters of $g_1$ in the neighborhood of those of $g_0$. Now, we can construct a transform $g_t$ with $0 < t < 1$ by interpolating between the group elements $g_0$ and $g_1$. We define thus a time collection of spatial group actions:

$$T_{g_{0\to 1}} = \{T_{g_t}; 0 < t < 1\}$$

Affine interpolation using the Lie group framework as been studied in [37]. This promising approach allows interpolation between $g_0$ and $g_1$ following the geodesic of the affine group. In our work, we resorted to a naive interpolation of the affine parameters : rotation angle, scale and shear.
We suggest to use a reasonable distance between $g_0$ and $g_1$ in order to make dynamic evolution of actions compatible with realistic anatomical motions in time dimension. Since this would lead to very small deformations in the case of brains, we consider that this dynamic extension of transforms should

be more relevant for non-rigid body imaging. We will only consider in our experiments the flip transform for dynamic EI. However, this kind of time indexed transform can be illustrated in the case of dynamic MRI data over an axis with the helical movement. If this movement isn't applying any transformation over the temporal axis, it uses this axis as a screw thread. If we progress over the given temporal axis, and slowly rotates the image of a few degrees and compute the interpolation accordingly, it results in a smooth helical movement that keeps the invariance property. It is a spatial transformation whose effect is operating over time axis. Angles used from one slice to another need to be very small




(a) Original volume

(b) Transformed volume

Figure 6.5: Helical Transform

## 6.3 Assessing Quality with Different Transformations

We aim to innovate from the previous researches led in MRI reconstruction with SSLEI using the temporal domain. We discuss the medical relevance regarding the use of anatomical data in Risk Analysis (7.5), but even if it might not be useful, studying the impact of temporal symmetry transformation is still of main interest.

To aware knowledge, no previous experiment have been led using SSLEI and dynamic data. The impact of the temporal symmetry to explore the null-space of $A$ isn't proved and might not give impressive results. This is why its efficiency can be assessed creating models with and without the temporal symmetry over the z-axis, and then compare both models quality. If the model only using combinations of flips, rotations and projections in the spatial domain, and without the temporal symmetry brings similar results, we can mitigate the relevance of the temporal symmetry.

We can also assess the impact of having only one transformation over an axis compared to multiple transformation over this same axis. Indeed, in our experimental study we "interpret" a third spatial dimension ($D$) of 3D volumes as a temporal dimension ($T$) leading to 2D+time data. Then, in the dynamic reconstruction view, we are only able to apply a single transformation over $T$, while in 3D reconstruction view, we can combine 3D rotations and flips.

It is not explored in this study due to limited computational resources, but if the temporal (2D+t) version brings poorer results compared to the 3D spatial version over the same slices sequence, we can assess the impact of getting only one transformation over a specific axis.

# Chapter 7

# Experimentation Environment

## 7.1 Objectives

From this part we experiment the different models discussed above from fixed-optimisation type algorithm to fully unrolled deep models involving an equivariant imaging loss in the context of self-supervised learning. All the models are first tested in 2D before tackling the challenging task of Dynamic MRI.

We first try to reproduce the results obtained in paper [4] regarding the use of equivariant imaging for 2D imaging in section 8. The main difference with the paper is the use of another dataset involving brain images with very complex structures that might not appear in the knee-fast-mri dataset. We will observe if the self-supervised models are able to reconstruct those structures with affine transformations.

Once we have convenient models in 2D, the most challenging part is to develop a 3D version of the current DeepInv library for Dynamic MRI. The addition of a new dimension reveals the hard task of memory management with limited computed power, and how to overcome it without altering the underlying properties of the physics in Dynamic MRI. Finally, we discuss the obtained results and some adjustments that can be made about the SURE loss, as well as possible further works that can be achieved regarding the evaluation of the equivariant losses in 3D self-supervised models.

## 7.2 Dataset

The implementation of the SSLEI for Dynamic MRI reconstruction is based on the QTIM dataset [7], which contains clinically-acquired anatomical and functional MRI scans. The 1.0.7 version of the dataset contains more than 1200 volumes already reconstructed with real and imaginary parts combined (only magnitude), without pre-processing of the acquisition noise. The Queensland Twin IMaging (QTIM) dataset (available here) contains anatomical and functional scans of 1026 young adult twins and siblings, aged between 18 and 30, as well as 176 twins aged 12 or 16. Although some participants were scanned four years later as part of the project, we only work on the first anatomical scan of each participant. This leads to a total database of 1202 NIfTI files, with 940 of them being of shape 256x256x256, 257 of shape 240x256x256, and 5 with different shapes. All of them are T1-weighted (T1w), a type of imaging commonly used for anatomical detail because they provide high contrast between different soft tissues. T1w images are particularly useful for identifying structures in the brain, spine, and musculoskeletal system. All scans are anonymous and therefore do not ethically impact the patients.

Even if the scans might come from different institutions, as they all capture the same anatomical sequence with a similar 4 T magnetic field, we neglect this data contributors' diversity's impact. Especially because the initial SSLEI proposed models are based on a PnP prior with a DnCNN, we expect our denoiser to be robust enough to handle the small acquisition differences. For better readability in the subsequent sections, the typical appearances of various tissues in T1w images are:

- White Matter: Appears bright.

- Gray Matter: Appears darker than white matter but brighter than cerebrospinal fluid (CSF).
- CSF: Appears dark.
- Fat: Appears bright.
- Bone (cortical bone): Appears dark.

Each patient (volume) also has a participant information sheet detailing specific acquisition parameters. The majority of the scans (no exhaustive checking was done) were acquired with an almost 4 Tesla (4T) magnetic field, which is considered high strength for MRI. This high field strength generally provides very good images due to increased Signal-to-Noise Ratio (SNR), improving image clarity and resolution compared to common clinical MRI scanners which typically operate at 1.5T or 3T.

The brain volumes were acquired directly in 3D, and therefore, need to be manually sliced to simulate the dynamic aspect. The slice thickness is 0.9 mm and the base resolution is 256, indicating that a typical acquired volume has a depth of 230.4 mm.

The acquisitions were performed with a partial Fourier factor of 0.875, meaning they were acquired with a fraction of k-space rather than the entire k-space. Consequently, the ground-truth data isn't fully-sampled. Additionally, a pixel bandwidth of 200 Hz/pixel was selected to minimize chemical shift artifacts and improve overall image fidelity, despite the increased noise. Addressing this noise is discussed in later sections. The given volumes only contain the module of the acquisition and not the phase part.

Note that on most of volumes, we can observe truncated parts of the patient due to head supports that help reducing the movement artifacts. Also, every file has a "defacing" version that we do not use (we want to work on the maximum number of data points to build a robust denoiser).

## 7.3 Requirements

As discussed in part 7, the implementation doesn't start from scratch. In order to achieve the project, most of the work is based on the Deep Inv library [16] which already contains a demo code for 2D self-supervised equivariant imaging learning with the fast-mri knee dataset and other demo codes for optimization algorithms. The library is composed of different modules that need to be articulated into a complete structure to achieve a specific task. We can already list the different components already implemented that we need to build our models:

- Dataset: dataset loader
- Denoisers: DnCNN (2D)
- Optimization procedures : PGD, ADMM, HQS...
- Optimization Priors: TV (2D), PnP...
- Training losses for inverse problems : MCloss, EI loss...
- Transforms: Affine transforms for EI (2D)
- Predefined imaging operators: 2D MRI physics
- Generators: subsampling mask generator (Gaussian...)
- Plotting: images and video plotting

In order to adapt the library to Dynamic MRI denoising, we need to add or adapt different existing modules:

- Dataset: QTIM dataset loader (for NIfTI files)
- Denoisers: DnCNN (3D), with batch-norm option
- Optimization Priors: TV (3D)
- Transforms: Affine transforms and flip time for Dynamic EI (2D+t)
- Predefined imaging operators: Dynamic MRI physics[1]
- Utils: patches to reduce memory usage (without altering physics)

## 7.4 Evaluation

### 7.4.1 Comparison Algorithms

In order to evaluate our reconstructed sequences, we compare our output to the ground-truth sequences as well as other denoising algorithms and optimisation procedures. They might use the ground-truth or not in the loss function instead of equivariant transformations to get knowledge about the null-space of $A$. We will only implement models with vanilla EI and REI, but no $\text{EI}_1$ or $\text{EI}_2$. Speaking about "SSLEI" models comprise both vanilla EI and REI models.

Our reference model we compare the SSLEI models with is a supervised unrolled PnP-ADMM, evaluated with Mean-Squared Error (MSE) metric. This is supposed to bring the best results compared to any other model as it learns a mapping between pairs of noisy measurements and ground-truth data.

We also implement a TV-ADMM ($\ell_1$-ADMM model-based) optimization algorithm to evaluate the relevance of learning a denoiser compared to applying a fixed constraint which is known to be efficient to denoise MRI images.

Finally, in order to assess the relevance of SSLEI with or without the temporal symmetry, tests are conducted with and without this flip (spatial transformations only). If the reconstructed sequences are almost the same as the ones incorporating the temporal symmetry, we can mitigate its relevance. In the same idea, building a 3D reconstruction model[2] might be interesting to study the impact of having only one transformation over the temporal dimension compared to multiple transformations in a 3D spatial domain (section 6.3).

### 7.4.2 Metric

Another aspect that needs to be explored is the metric used to evaluate the different reconstruction algorithms. As we are working on dynamic MRI data, the famous and commonly used Peak Signal-to-Noise (PSNR) metric which is designed for static images might not really represent accurately the reconstruction quality.

The PSNR is a metric commonly used in image reconstruction to evaluate a model quality. The PSNR measures the ratio between the maximum possible power of a signal and the power of corrupting noise that affects the fidelity of its representation.

$$\text{PSNR} = 10 \cdot \log_{10} \left( \frac{\text{MAX}^2}{\text{MSE}} \right) \tag{7.1}$$

where "MAX" is the maximum possible pixel value of the image and "MSE" is the Mean Squared Error between the original and the altered image.

[1] During the implementation period, an update of Deep Inv on 10/06/2024 added the physics for DynamicMRI that we used afterward. However, some parts were missing and we had to edit it.

[2] We don't assess this aspect in the research. It would require very expensive resources to work on non-patches 3D volumes.

In the case of static reconstruction in the image domain only such as 2D-CT, it's very easy to use and implement. However, we are working with two-channels Dynamic MRI data due to the real and imaginary parts. We therefore need a complex PSNR metric or to compute the PSNR on normalized channels.

Also, as we are not in the case of a real functional MRI reconstruction (we flatten a volume, we don't work on the same slice over time), such a metric can be biased. If the PSNR can also be applied to 3D volumes as it consists of a norm computation, the dynamic range of pixel values in dynamic MRI sequences may vary across frames due to changes in contrast or intensity over time. This is why in areas with few features, such as at beginning and end of the volume which are all black, we can easily denoise and reconstruct them, highly increasing the PSNR value. But in areas with more features (middle of the volume), very complex structures are harder to reconstruct but are the purpose of the study.

We do not implement it, but adding a convex-hull around the brain and compute the complex PSNR over this area would be more precise.

We can therefore obtain a 3D complex PSNR but this index might be biased. Another approach could be to compute the complex PSNR for each frame of the volumes, resulting in a time-varying PSNR over the temporal axis. This would layer more precise information about the reconstruction quality in different parts of the volume.

As the PSNR is expressed in decibels (dB), the scale is logarithmic. This is why a small variation in PSNR might represents significant change in denoising. Mathematically, if the initial PSNR is $\text{PSNR}_1$ and the final PSNR after a 3 dB decrease is $\text{PSNR}_2$, then:

$$\text{PSNR}_2 = \text{PSNR}_1 - 3$$

Since:

$$\text{PSNR} = 10 \cdot \log_{10}\left(\frac{\text{MAX}^2}{\text{MSE}}\right)$$

we have:

$$10 \cdot \log_{10}\left(\frac{\text{MAX}^2}{\text{MSE}_2}\right) = 10 \cdot \log_{10}\left(\frac{\text{MAX}^2}{\text{MSE}_1}\right) - 3$$

Dividing both sides by 10:

$$\log_{10}\left(\frac{\text{MAX}^2}{\text{MSE}_2}\right) = \log_{10}\left(\frac{\text{MAX}^2}{\text{MSE}_1}\right) - 0.3$$

Using properties of logarithms:

$$\log_{10}\left(\frac{\text{MAX}^2}{\text{MSE}_2}\right) = \log_{10}\left(\frac{\text{MAX}^2}{\text{MSE}_1 \cdot 10^{0.3}}\right)$$

This implies:

$$\text{MSE}_2 = \text{MSE}_1 \cdot 10^{0.3}$$

Since $10^{0.3} \approx 2$, we find:

$$\text{MSE}_2 \approx 2 \cdot \text{MSE}_1$$

Thus, a decrease of 3 dB in PSNR means that the noise power has doubled.

## 7.5 Risk Analysis

The proposed research and experiment are based on 3D volumes that we transform in order to simulate Dynamic MRI data. Manual slicing and degradation of MRI volumes for simulation purposes offer the advantage of controlling factors such as noise and acceleration factor. However, the resulting simulated data may not fully capture the complexities and variability present in "real" dynamic MRI data. Since we are not working on "real" dynamic data, the experiment mostly gives an idea about the potential of SSLEI to reconstruct dynamic MRI sequences but is limited in its clinical adoption. The absence of easy access to dynamic MRI data also comes from the limited usage of such type of MR imaging in clinic.

In addition, when we want to image a temporal evolution in MRI, instead of dynamic anatomical MRI sequences, we prefer fMRI to study blood circulation (see section 3.3). Its purpose is to study the blood nature and how it irrigates the brain during a specific cognitive task. This data is of low spatial resolution and inherently becomes interesting in a temporal aspect. At the moment, we can't expect to obtain these "high" resolution fMRI sequences for professional usage. The current fMRI reconstructed images result in brain images with a drooling stain. In short, this first approach of the Dynamic MRI reconstruction might be a step towards reconstructing quality fMRI images, which layers more interest regarding the dynamical aspect, but only if the experiment shows robustness even for very bad resolution and noisy sequences.

Also, the model training can be expensive at full resolution. We face GPU limitations and computation time limitations. Just for 2D MRI reconstruction with 900 images and a resolution of 320x320, the model trained for ≈12h with author's hardware and a batch size of 2. Therefore, in the case of dynamic data at 256x256x160 sequences (we crop time axis to avoid non-relevant areas), the training might last more than a day to obtain a convenient result, which is extremely expensive, and we will surely be limited to a batch size of 1. In addition, with unrolling algorithms that aim to learn a denoiser, the memory resources (e.g. GPUs) can quickly become saturated. There is a hardware limitation to such a model. Finally, each built model is for a specific degradation (e.g. acceleration factor = 4, noise sigma = 0.1) and each new degradation requires a new learning. Therefore, as lot of parameters can be tuned for MRI acquisition (multi coil, acceleration factor...), and as the acquisition itself can fluctuate according to each MRI machine, having a general model for any MRI machine is not possible yet.

## 7.6 PLES issues

Although we aim to develop an experiment that is robust for medical imaging, they are some limitations we need to consider that can impact the future results reading and/or utilisation.

On a professional aspect, they are obvious clinical limitations to such a project. Firstly, MRI data acquisition is often more complex than the simple linear operator $A$ we considered during this experiment. This transformation matrix may include the influence of the excitation pulse sequence, the form of k-space sampling and uncertainties like the coil sensitivity. If the transform $A$ is not linear, this might brings worse results (even if the initial idea extends to more general operators). Also, as mentioned earlier, the interest of dynamic MRI for T1, T2,... sequences might be limited, but is a first step towards fMRI reconstruction which is inherently dynamic.

Regarding legal, ethical and social (LES) issues, this is exclusively linked to the database. In medical imagery, and in data-driven models in general, we must be vigilant about the data we have. Especially, a database should not over-represent a specific ethnic group or sex. This is for the generalisation and fairness of the model, as anatomic differences can lead the model to be "adapted" to a specific group of people. Under-representation of populations in clinical trials has been highlighted in numerous studies [38] with tests that mostly represent white males. Here, the particularity of the dataset having twins participants is ignored. The QTIM dataset is assumed to be built based on ethical compliance, based on their statement: "Written, informed consent was obtained from all participants, including a parent

or guardian for those aged under 18 years, and the study was approved by the Human Research Ethics Committee at the QIMR Berghofer Medical Research Institute".

If the issues would be linked to the database, our algorithms should not cause any issue as they intrinsically do not discriminate any ethnic group. As we didn't acquire our own data, it relies on the considerations taken by the QTIM creators and shares to openly distribute ethical data (we expect it to be correct). We rely on the ethical work done by the data contributors to stitch to the different ethical and socials issues. Our implementation only operates on the de-identified patients brains pixels. Regarding the legal aspect alone, every publication and results based on this data is for non-commercial purposes only.

## 7.7 Software and Hardware

All the codes in this project are written in Python to be grafted on the pre-existing Deep Inv library and possibly be added to the library. This library is built on PyTorch modules and therefore require an environment for PyTorch.
As there is the major transition to 3D, the memory usage in this experimentation is way bigger than simple 2D models. In order to train and test the different algorithms without expensive calculation time, we use CUDA (11.8 or 12.1) to use the GPU fast computation. However, to avoid saturating the GPU memory too fast, we will use diverse tricks (see [8.1.2, 9.1.2, 9.2] for details) to learn full-resolution models, and we will load and degrade the ground-truth dataset with the CPU to get the pairs of ground-truth / measurements.
For this project we use a simple NVIDIA RTX3050 laptop GPU with 4 GB of memory in a CUDA 12.1 environment for 2D models and fixed parameters models, but a bigger station incorporating a RTX 3080 with 10 GB of memory in a CUDA 11.8 environment for unrolled 3D models. This card has 8704 cuda cores allowing fast tensor computation.
The debugging and tests of low resolution are first developed on the laptop, and the more powerful station is used only to launch train the models in full-resolution and more epochs to get the best results.

## 7.8 Data Preparation

### 7.8.1 Denoising the original dataset

When we start working with the original volumes from the QTIM dataset, we quickly face two difficulties to evaluate the model quality. First is the wide range of values for each pixel/voxel for the different patients, with few values reaching +450 and other going under 0. To avoid this type of outliers, instead of a min-max scaling over the whole range of intensities, that would bias our volumes, we apply a min-max scaling in function of the 99.5th percentile. Secondly, if the PSNR is very interesting to estimate the reconstruction quality of a model compared to the original sequence, it requires this original sequence to be a reliable ground-truth. However, the volumes acquired in the QTIM dataset are not cleaned and lot of noise degrades the volumes quality. Therefore, when adding more degradation to the volumes (acceleration factor and noise in k-space), when we try to simply apply a TV-denoising to the slice, if the result is visually convenient, the PSNR is indicating a loss of quality after the denoising. This is because the ground-truth is worse than the reconstruction! On the long term, it appears that it is not acceptable to keep noisy ground-truth as a comparison point for our models. Also, it is hard to distinguish the additional noise from the original noise. To deal with this issue, there is no other solution than to already denoise the database itself. To process a volume, it appears that a combination of TV-denoising followed by a $\ell_1$-denoising is efficient. The TV-denoising removes the parasite noise inside the brain volume resulting in anisotropic diffusion and edge-preservation, while $\ell_1$ denoising is good to remove the diffused redistributed noise by promoting sparsity in the denoising process. This second denoiser is desired to have flat "zero" areas around the volume. The objective is that if we apply a convex-hull over the denoised volume, it would easily find the brain structure itself without any perturbation from the noise in the black areas due to MRI acquisition process (Rician noise). For reproducibility, we use:

Table 7.1: Denoising Parameters

Table 7.2: Parameters for TV-Denoising

| **Parameter** | **Value** |
|---|---|
| Maximum Iterations | 50 |
| Step Size | 0.5 |
| Regularization Parameter ($\lambda$) | 0.02 |
| Maximum Iterations | 25 |

Table 7.3: Parameters for $\ell_1$-Denoising

| **Parameter** | **Value** |
|---|---|
| Step Size | 0.25 |
| Regularization Parameter ($\lambda$) | 0.1 |
| Maximum Iterations | 50 |

This leads to denoised volumes that are almost noise-free, meaning we can use them as trustful ground-truth to evaluate our models (despite the partial Fourier factor of 0.875). The noise we add by degrading the volumes is not mixed with the original noise anymore. However, let's keep in mind that this suggests that we alter the original database for the project.

### 7.8.2 Handling the NIfTI files

To facilitate working with tensors in the commonly accepted format used by the PyTorch library —(B, C, H, W) for 2D data and (B, C, D, H, W) for 3D data— we develop a specialized class to fit the Deep Inv library. This class is designed to convert NIfTI data files into tensors, accommodating both 2D and 3D data formats.

The class takes several inputs: the dataset path, which specifies the location of the NIfTI files; the data dimension, indicating whether the data in the dataset is 2D or 3D; a downscaling factor, which allows for reducing the data resolution if needed; and the slice position, which enables the extraction of specific 2D slices from a 3D dataset if only 2D analysis is desired.

The class is built to handle resolutions that are powers of 2 (and divide 256), ensuring that the data is compatible with deep learning practices and minimizing information loss through average pooling. It performs the percentile min-max scaling to normalize the data and reshapes it to the desired dimensions —either 2D or 3D— based on user specifications. The processed data is then converted into tensors with the formats (B, C, H, W) for 2D data and (B, C, D, H, W) for 3D data, and is returned as float32 tensors to optimize memory usage (64 doubles the memory space). In the case of dynamic 2D data, the depth (D) dimension may represent the time (T) dimension.

By default, the class utilizes two channels for MRI data, corresponding to magnitude and phase (real and imaginary parts), and loads all the files. In this study, we keep two channels for MRI even if the data is only contained in the first channel (discussed in 10). However you can set a specific number of channels and maximum number of files to take. This design accommodates a range of user needs, from those requiring full-resolution 3D data to those with limited computational resources who need smaller dimensions.

This class is an almost ready-to-merge class to the Deep Inv library but might necessitate finer adjustments and always requires the user to specify both strategy to get sub-folders and strategy to get files inside the sub-folders. Notice that this class only load and returns the whole batch of files and do not save them in a dataset. In order to save the ground-truth files we also added our own "datagenerator" that only deals with ground-truth files and store them in h5 format. This is derived from the initial datagenerator in Deep Inv.

# Chapter 8

# 2D Models

In this section we aim to reproduce the results obtained by Dr. Chen, J. Tachella and M.E. Davis in their paper about equivariant imaging [4]. As this study has already been proposed and is not the aim of our research, we implement the different reconstruction models with only one fixed acceleration factor $R = 4$ and one noise level $\sigma = 0.05$. The exact same subsampling is applied for every architectures. $\sigma = 0.05$ should be a good compromise between a noise-free image and relatively noisy image.

When the model is trainable (i.e. when we use a DnCNN), we take 400 images for training and 100 images for evaluation. Results are averaged over the test set. We use a complex PSNR index despite the original data being in only one channel (magnitude only). The relevance of this PSNR index is discussed in 10.

For reproducibility, we always extract the slice situated at 52% of the depth of each volume.

For reference, the linear reconstruction over the test set has an average complex PSNR of 22.7±0.5 dB (computed over the 100 images) at $\sigma = 0.05$.

## 8.1 2D Fixed Parameters Models

### 8.1.1 2D TV-ADMM

Using the different elements provided in Deep Inv we can easily build a model for a 2D TV-ADMM reconstruction. This method can be interpreted as a direct concurrent of self-supervised learning with equivariant imaging as the ground-truth is never used to learn the reconstruction, but however requires fixed parameters. For reproducibility:

Table 8.1: Parameters for 2D TV-ADMM

| **Parameter** | **Value** |
|---|---|
| TV Prior Maximum Iterations | 150 |
| Step Size | 0.6 |
| Regularization Parameter ($\lambda$) | 0.025 |
| Maximum Iterations | 150 |

Parameters can still be adjusted to lead to slightly better results. Using the complex PSNR, over 100 test images we reach 26.3±0.6 dB on average. This results in a gain i of $\approx 3.6$ dB compared to linear reconstruction. As this indicator relevance can be discussed, we should also consider the visual aspect. Over a single single brain image:

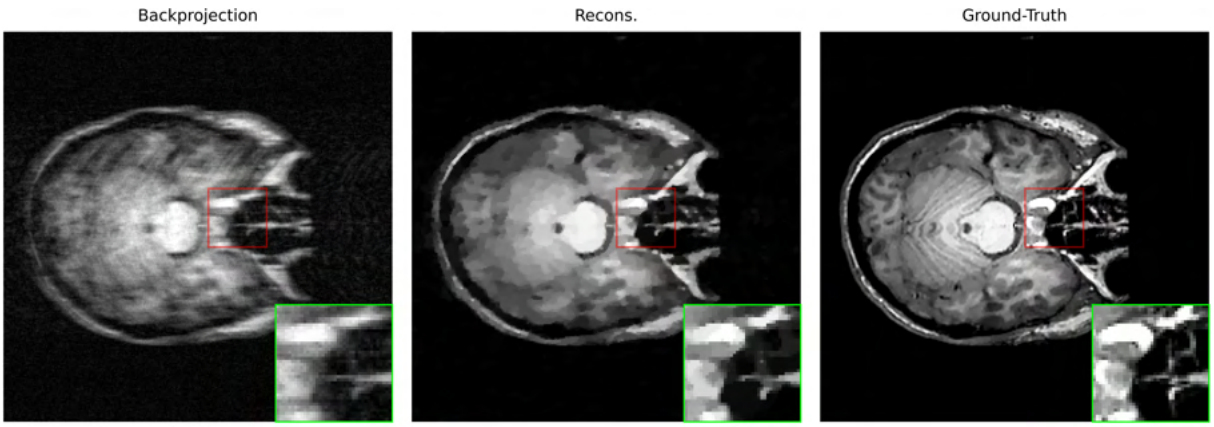
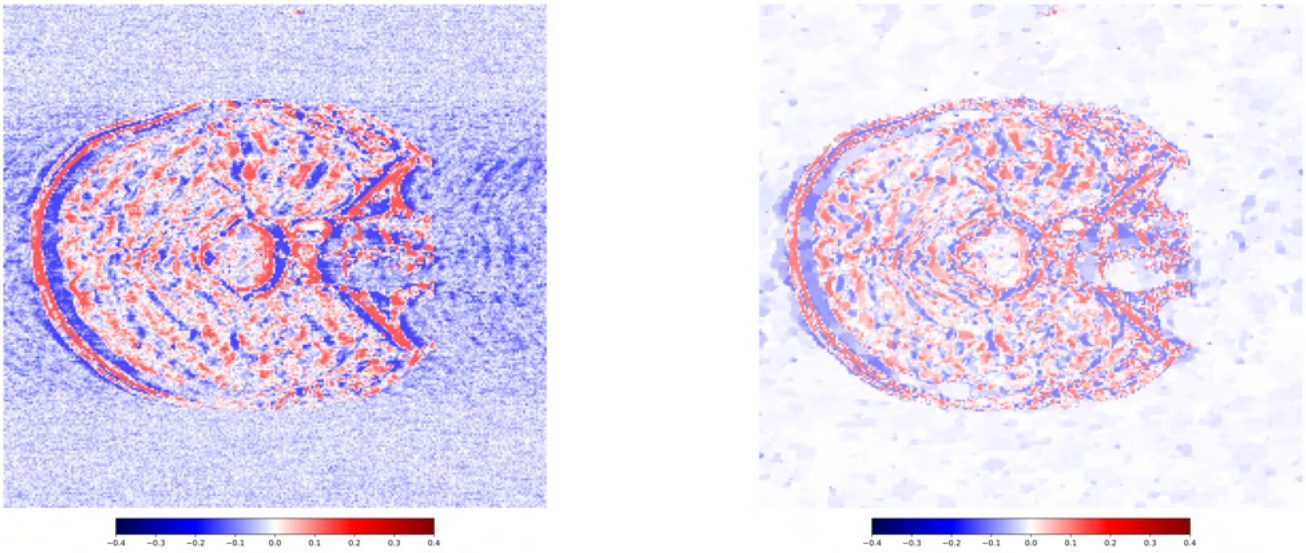



Figure 8.1: 2D TV-ADMM Reconstruction

We also visually observe an improvement compared to the linear reconstruction with acceleration artifacts partially corrected. It's already a relatively convincing result regarding the absence of ground-truth in the optimization process. The image is still somewhat blurred with a loss of edges and no fine details. The error is computed doing ($error = ground\text{-}truth - reconstruction$), bounded between $\pm 0.4$ and clipped to not exceed this range, explaining saturation in backprojection.



(a) Linear Error

(b) Reconstruction Error

Figure 8.2: Error in 2D TV-ADMM Model

### 8.1.2 2D PnP-ADMM fixed DnCNN

In a fixed parameters denoising architecture of type PnP-ADMM, we have to learn a denoiser before applying the iterative optimization. The denoiser we train is a DnCNN learning a mapping between ground-truth images and noisy images. We do not learn the mapping in the Fourier Space but can chose to degrade either in the image domain by adding white Gaussian noise, or in the k-space resulting in Rician noise in the image space. We do not apply any acceleration (mask) in this mapping as it is the role of the iterative optimization procedure (here ADMM) to correct the acceleration in the k-space. Both models use the Denoising physics in the training, but are degraded with different strategies.
As processing 256x256 images might be expensive depending on the resources, and as authors did in [2], working in the image space only allows the use of patches to reduce the memory usage especially while working with the GPU. Using a patch of size 32x32, assuming there is no overlapping, we can use (256x256)/(32x32) = 64 patches for each image.

A DnCNN trained over smaller patches can also be applied over the whole image and does not alter the deployment over the 256x256 image [2]. We therefore use in this example 32x32 patches for training but the 256x256 images during evaluation and tests. We parameterise for both DnCNNs:

Table 8.2: Parameters for 2D DnCNN

| **DnCNN config** | **Value** |
|---|---|
| Batch Normalization | False |
| Depth | 17 |
| Number of Filters (nf) | 64 |
| **Training Parameters** | |
| Epochs | 10 |
| Learning Rate | 3e-4 |

We first add the noise to the images before cropping 400x64 (25600) patches. For the DnCNN with the WGN in the image space we use the "Denoising" physics instead of the MRI physics (no MRI DnCNN). We apply the same strategy for the second DnCNN but with the noise directly added in the k-space.

We use a train batch size of 48, and of 16 for eval. We train over 10 epochs only and we use a multi step LR scheduler at epochs 4 and 8, with $\gamma = 0.25$. Scheduler might not be necessary. After 10 epochs for each DnCNN training, the different 32x32 patches have been denoised such as:

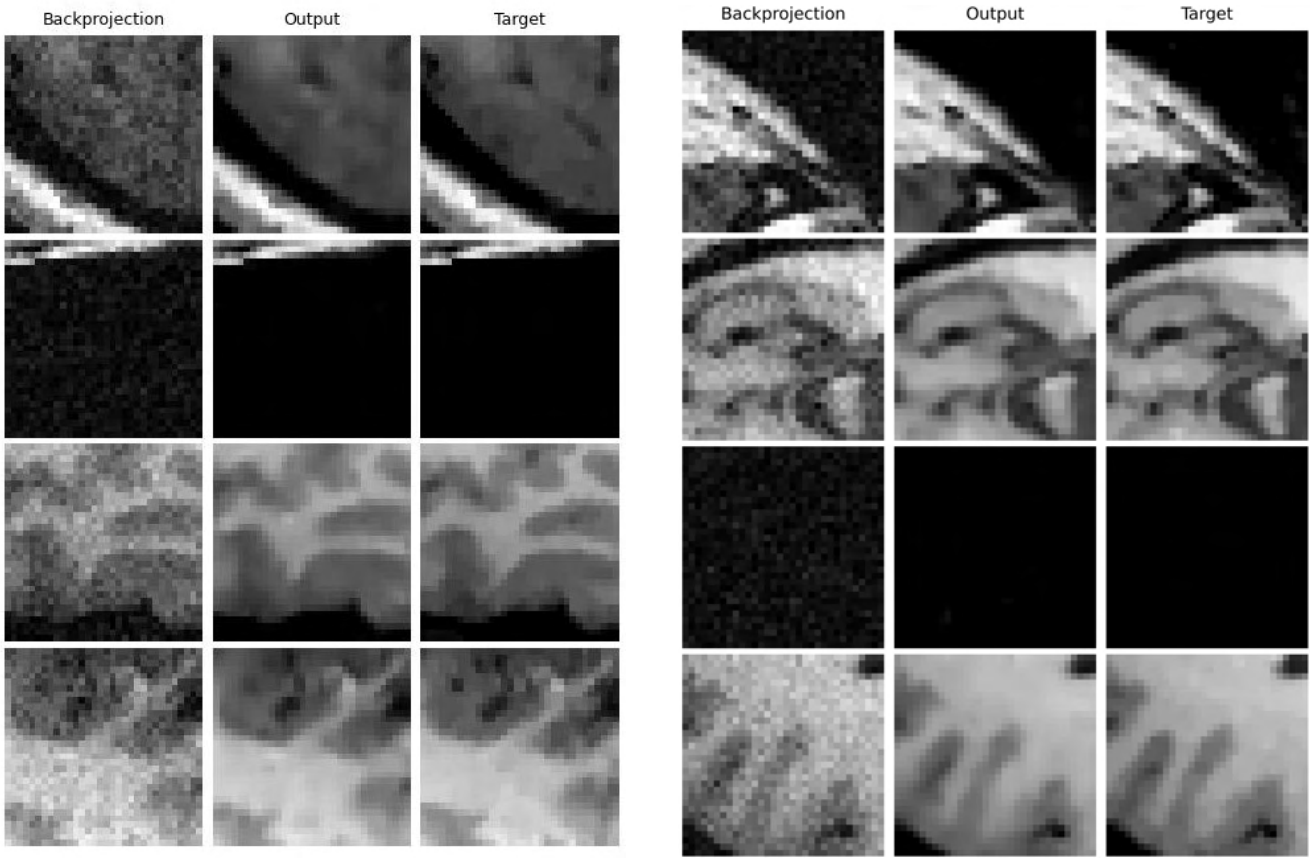



Figure 8.3: PnP-ADMM Denoised Patches. left: WGN in image space, right: WGN in k-space

The mapping excels in denoising but suffer from a small overfitting. For both DnCNNs we have an original complex PSNR of 23.9±0.1 dB. Over 100 images, we obtain a PSNR of 34.5±0.5 for WGN in image space and 34.4±0.4 for Rician noise version. Both results are very comparable.

For both optimizations with the fixed DnCNN we use the model parameters:

Table 8.3: Parameters for fixed 2D PnP-ADMM

| **ADMM Parameters** | **Value** |
|---|---|
| stepsize | 1.75 |
| beta ($\beta$) | 1.25 |
| max iter | 50 |

### DnCNN with noise in image space

Incorporating the pre-trained DnCNN in the PnP-ADMM denoising architecture, we obtain a complex PSNR of 28.7±0.7 over the test set. It's an improvement of 6 dB compared to backprojection.

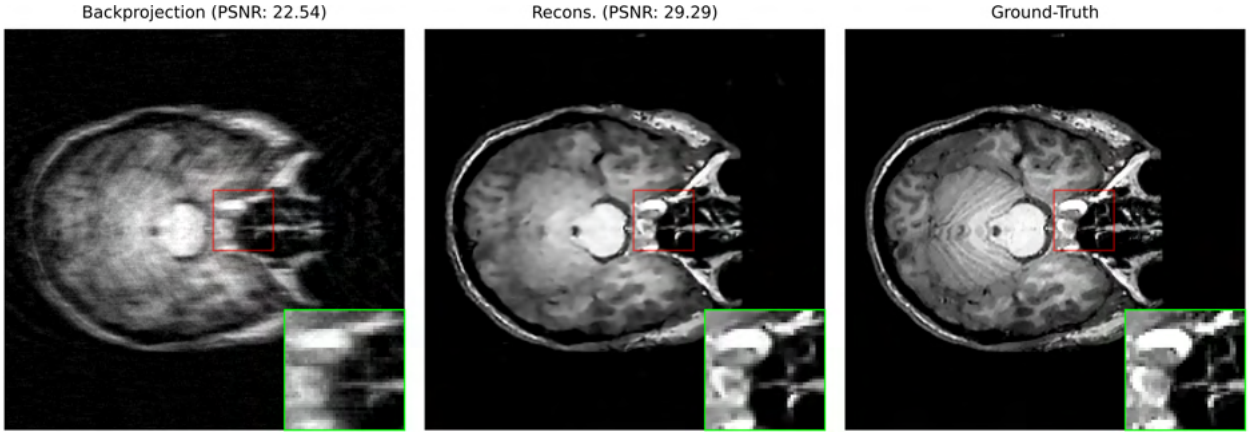
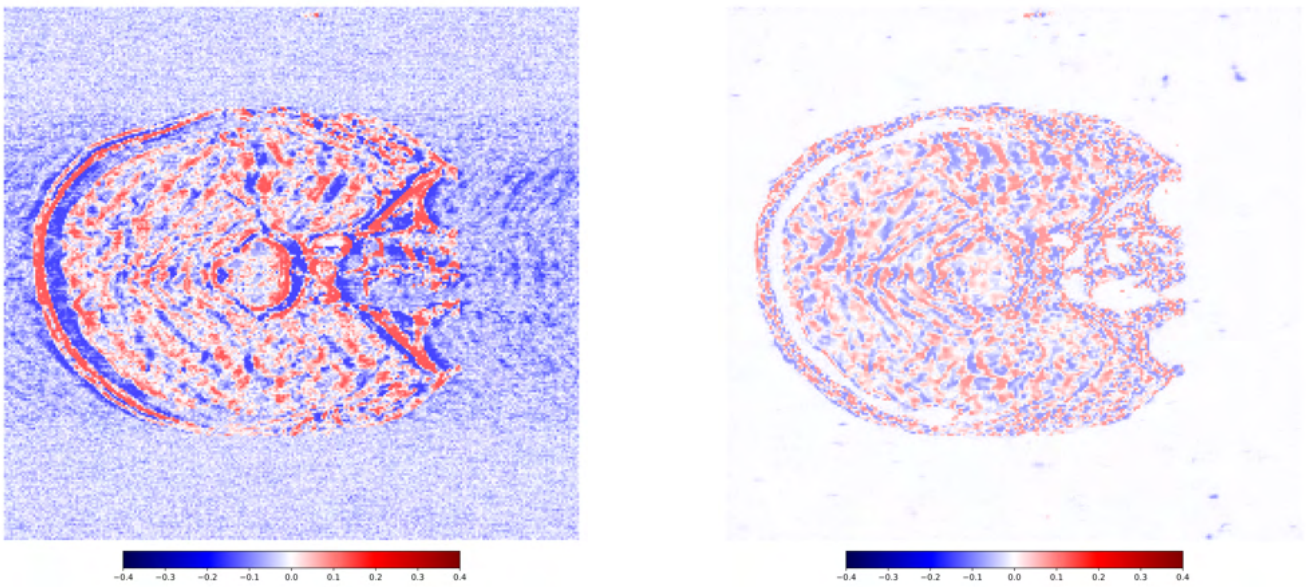



Figure 8.4: 2D PnP-ADMM Reconstruction WGN (PSNR values are computed over this image only)

On this specific image, the reconstruction is considerably better than the input with a gain of ≈ 6.8 dB. The acceleration effect is corrected but the model tends to blur fine details areas too much.



(a) Linear Error

(b) Reconstruction Error

Figure 8.5: Error in 2D No-MRI fixed DnCNN PnP-ADMM Model

**DnCNN with noise in Fourier space**

Applying the PnP-ADMM routine with noise in Fourier space over the same brain results in a PSNR of 28.7±0.7, which is equal to the previous model.

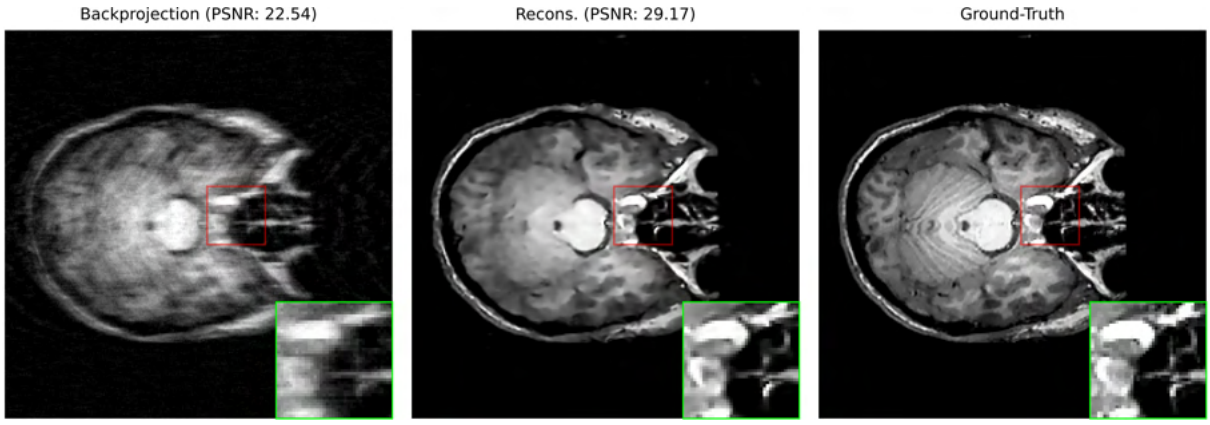
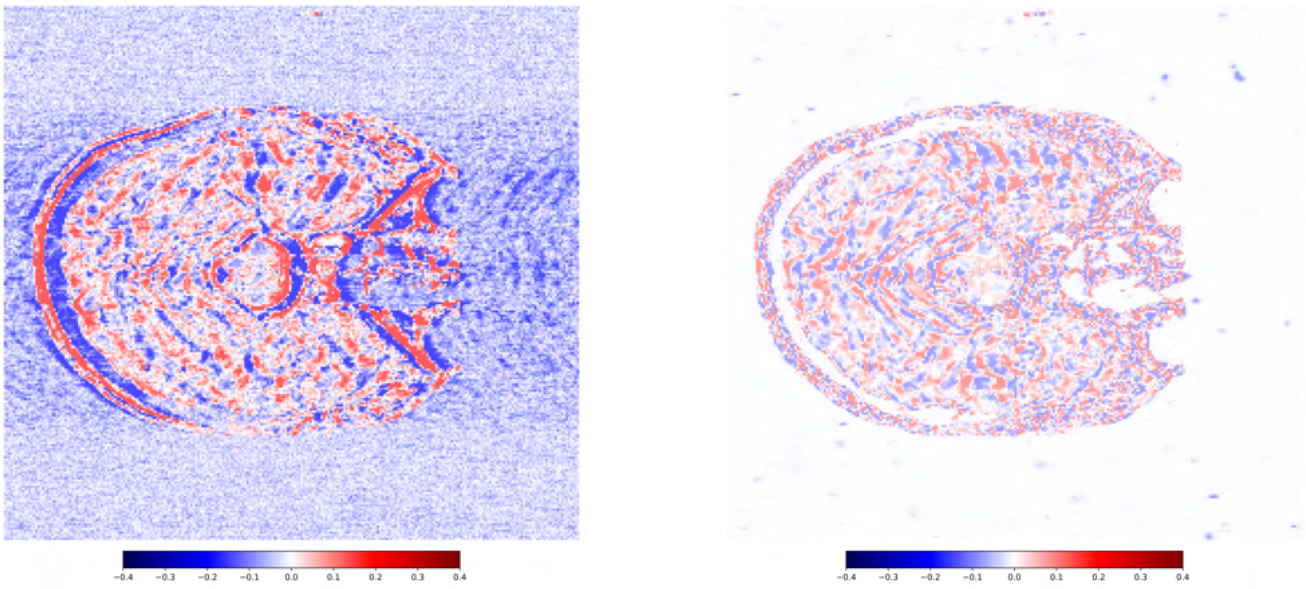



Figure 8.6: 2D PnP-ADMM Reconstruction

We obtain a very similar image with equal PSNR at $10^{-1}$ precision.



(a) Linear Error

(b) Reconstruction Error

Figure 8.7: Error in 2D MRI fixed DnCNN PnP-ADMM Model

Both pre-trained DnCNN models surpass the TV-ADMM reconstruction. Especially in the inset, the fine structure is accurately reconstructed. This improvement compared to TV can be explained by the DnCNN having learned a mapping between ground-truth and noisy images. However finding the good parameters for the ADMM procedure is time consuming and might differ from one patient to another to obtain better results.

## 8.2 2D Unrolled Models

In unrolled architectures, instead of passing a fixed DnCNN we have the ambition to learn it progressively over a certain number of iterations. Also, the *stepsize* and $\beta$ parameters of the ADMM can be optimized along with the denoiser learning. Patching in this architecture is harder because we need to patch data

for the DnCNN inside the Trainer, so we keep images in full resolution. Due to the unrolled architecture, we reduce the denoiser architecture and iterator iterations.
Compared to [1, 4, 6], we do not use a U-Net but a DnCNN, and we apply multiple iterations of the Proximal operator (not only 1). We use a StepLR scheduler operating after 120 epochs (80% of 150). For every unrolled models we set:

Table 8.4: Parameters for 2D unrolled PnP-ADMM

| **DnCNN config** | **Value** |
|---|---|
| Batch Normalization | False |
| Bias | True |
| Depth | 7 |
| Number of Filters (nf) | 64 |
| **Iterator Parameters** | |
| iterator | ADMM |
| iter | 3 |
| **Training Parameters** | |
| Epochs | 150 |
| Learning Rate | 5e-4 |

The ADMM parameters are lists of length "iterator iter" (so 3 here) and initialized at $stepsize = 1.0$, $\beta = 1.0$ and $gparam = 0.1$, but are progressively refined by the unrolled architecture.

Note that in this section we use the lowest loss epoch of the training to compute the metrics because observed that the loss can suddenly fluctuate.

### 8.2.1 2D unrolled supervised PnP-ADMM

The supervised unrolled model will serve as a reference to what "we can aim to reach" with the unsupervised models. With the supervised loss (MSE metric), we reach a PSNR of 28.8±0.8 dB. The PSNR value over the test set for the linear reconstruction being of 22.7±0.5 dB, we have a gain of ≈ 6 dB. The result obtained is very closed to what we obtained with the pre-trained DnCNN !

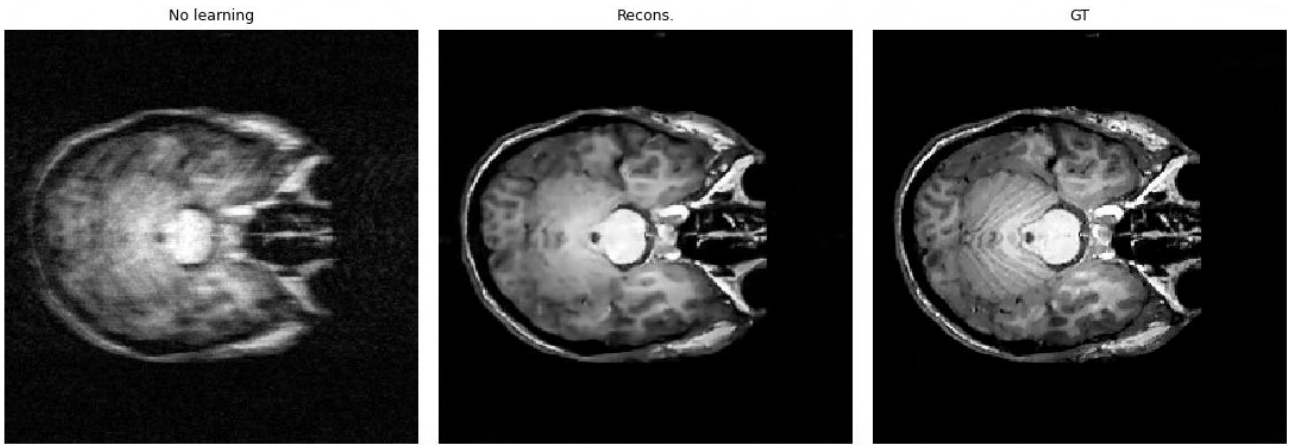
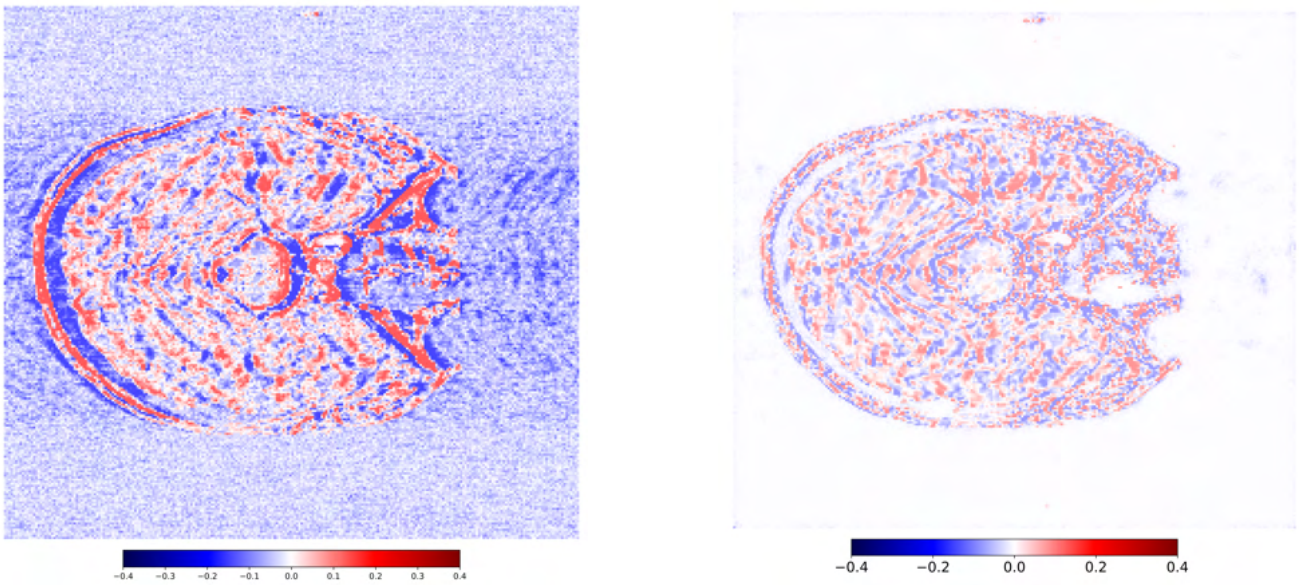



Figure 8.8: 2D Supervised Reconstruction

Visually, compared to the pre-trained DnCNN model, it is just a bit sharper and slighly better in fine details structures. It's the best model at denoising the noise outside the brain, with very few and small persistent noise.




(a) Linear Error

(b) Reconstruction Error

Figure 8.9: Error in 2D Supervised Model

In [6], with a noise of $\sigma = 0.05$, adjoint reconstruction had a PSNR of 24.7 and supervised model resulted in a PSNR of 30.8 dB, suggesting a gain of $\approx 6.1$ dB. We obtain very similar results with a gain of $\approx 6$ dB. The difference in PSNR can be explained by the different nature of MRI images, with here an influence from the empty black areas (no features areas) and the complex structures areas (discussed in 10). What matters is to conserve the rough size difference between the supervised models and unsupervised models, to demonstrate reproducibility.

### 8.2.2 2D unrolled unsupervised PnP-ADMM EI

Regarding the unsupervised model with equivariant imaging loss, [6] showed that the noise level quickly impact the reconstruction quality as $\sigma$ is increasing. Going from $\sigma = 0.01$ to $\sigma = 0.05$ implies a loss of 3 dB with the knee MRI dataset, from 30.6 to 27.6 dB. Therefore, we expect a PSNR of $\approx 25.7$ to be a coherent result. For our training, we use affine linear transformations detailed in [5] in the EI loss and a simple MC loss for the data fidelity. The model has to find a balance to minimize the total loss, while converging towards a visually convenient solution. In [6], authors use $\alpha = 1$ for MRI. We use the Affine group of transformations with small modifications: we suppress the shift (parameter set to 0) and the minimal zoom factor for scaling is set to 0.8.

Table 8.5: Parameters for 2D EI loss

| **EI Loss Parameters** | |
|---|---|
| Transformations | Affine |
| Number of transformations | 1 |
| No Grad | True |
| Apply Noise | False |
| Weight EI loss ($\alpha$) | 1.0 |

After training, we only obtain 24.0±0.5 dB which is only 1.3 dB more than the backprojection. It is very low result regarding what we were expecting (we expected 1.7 more decibels of difference).

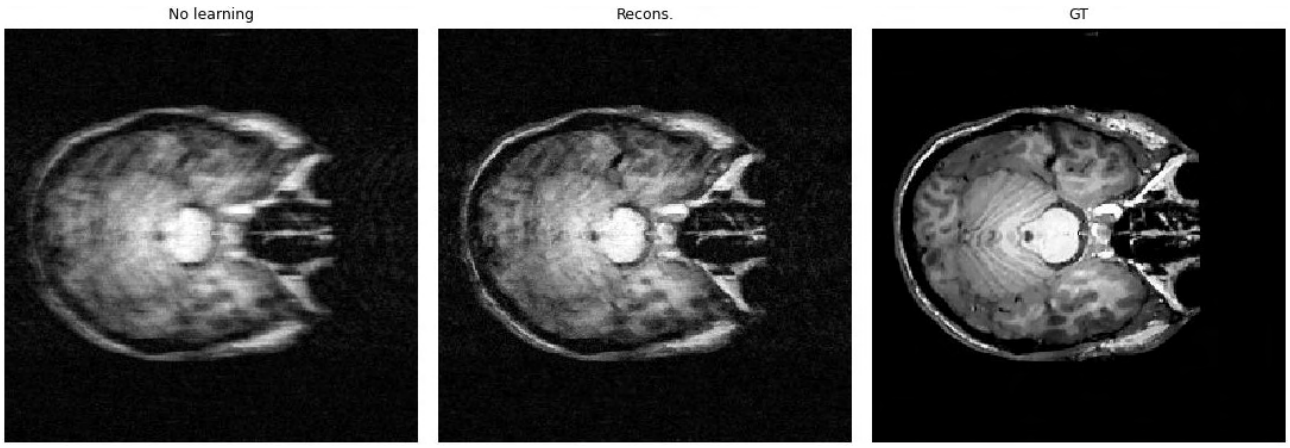
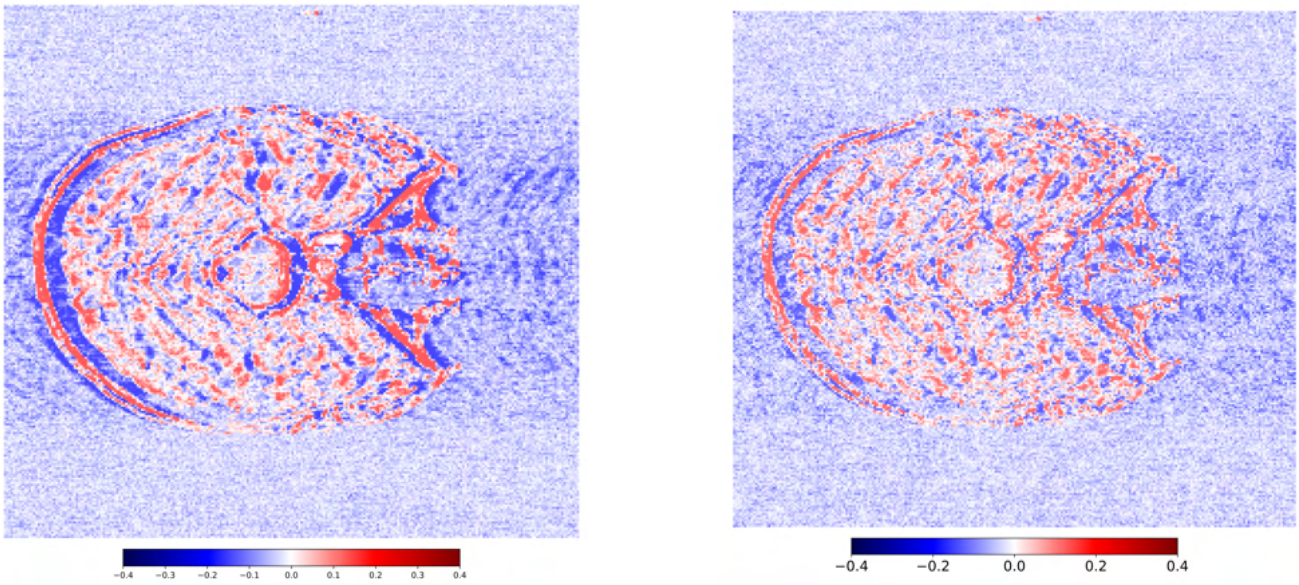



Figure 8.10: 2D Unsupervised EI

Visually, it's probably the worst result with no fine structures in the reconstruction. Some edges are just a bit sharper, but acceleration artifacts and noise is still very strong. We can't exploit these images.




(a) Linear Error

(b) Reconstruction Error

Figure 8.11: Error in 2D vanilla EI Model

### 8.2.3 2D unrolled unsupervised PnP-ADMM REI

In their paper [6], authors experimented the robust equivariant imaging with a SURE Gaussian loss (see 5.3.2) for data fidelity coupled with noise handling in the EI loss. At a noise level of 0.05 $\sigma$ they observed a loss of 1.8 dB compared to the supervised model, which is 2 dB better than the initial EI model. This suggests we should obtain a PSNR of $\approx$ 26 dB if the behavior is the same.

Table 8.6: Parameters for 2D REI loss

| **EI Loss Parameters** | |
|---|---|
| Number of transformations | 1 |
| No Grad | True |
| Apply Noise | True |

Using similar parameters for the EI loss but this time using the "apply noise" option joint with a SURE Gaussian loss for the data fidelity term, we reach a PSNR of 28.0±0.7 dB. This overpass our expectations with a gain of 4 dB compared to the simple EI model, and being way closer to the supervised model results (0.8 dB of difference).

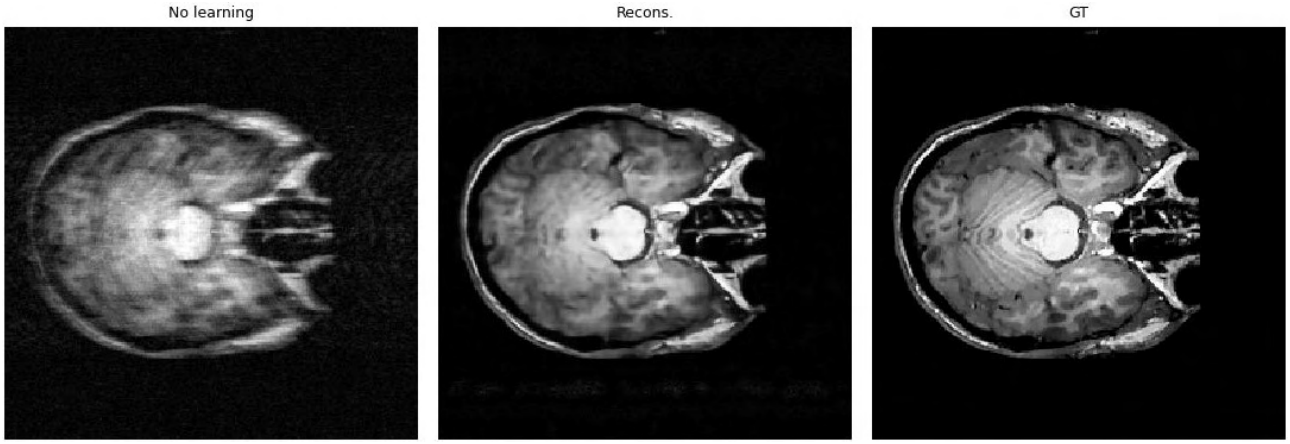
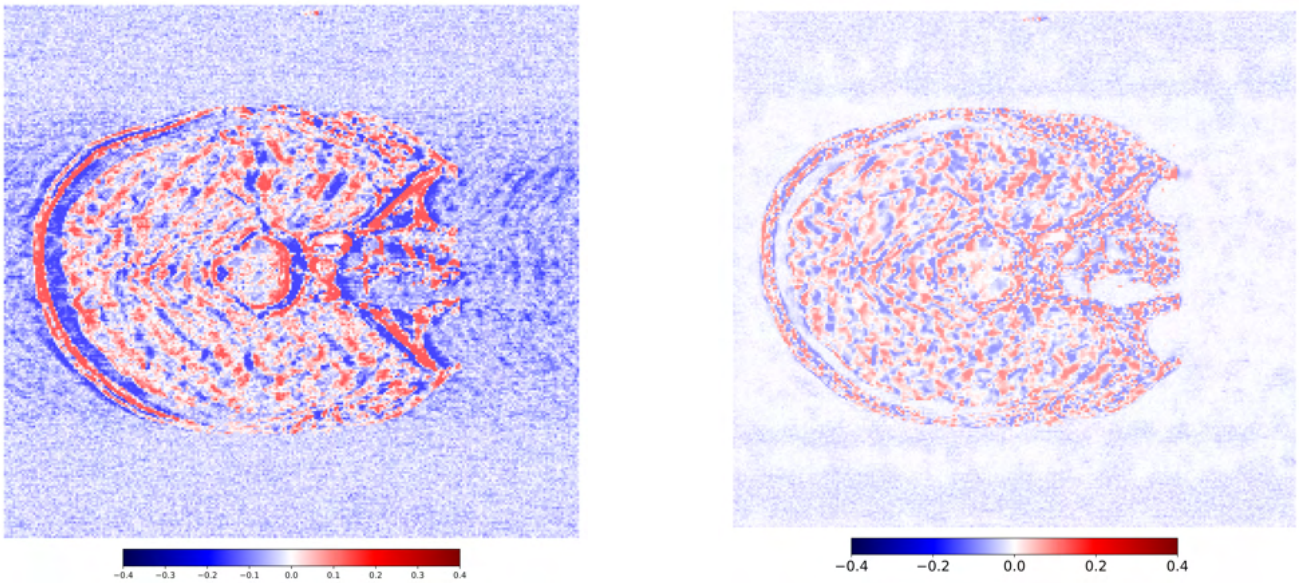



Figure 8.12: 2D Unsupervised REI

Visually, it's probably the most convincing result with fine structures that are reconstructed by the REI model but were not reconstructed by the supervised model. Even if the PSNR is lower than the fixed PnP-ADMM, it can be prefered to exploit for clinic usage. We can still nuance the reconstruction quality at edges due to the borders between grey and white matter being too smooth.
Also, we surprisingly accurately denoise the Rician noise in the central band (we define the central band the part with visible acceleration artifacts) but we fail to suppress it in the upper and lower bands. In other words, REI worked very well in areas containing features but had mitigated results in empty areas. It is even more surprising when we notice that the original noise is weaker in these two bands than in the central band, where the acceleration also adds perturbations.




(a) Linear Error

(b) Reconstruction Error

Figure 8.13: Error in 2D REI Model

## 8.3 2D Comparison, comments and further work

### 8.3.1 Comparison

| $\sigma$ | $A^\dagger$ | TV-ADMM | PnP-ADMM[1] | EI | REI | Supervised |
|---|---|---|---|---|---|---|
| 0.05 | 22.7±0.5 | 26.2±0.6 | 28.7±0.7 | 24.0±0.5 | 28.0±0.7 | 28.8±0.8 |

Table 8.7: PSNR Comparison of 2D models

Globally, we observed lower PSNR values but for every model compared to [6]. These lower PSNR values, compared to what was observed in the original paper, can be explained by the different architecture of the models, a different denoiser, and the different type of images. Here, neuroimages contain more complex structures but also dark empty areas without features. What matters is the difference between the different models.

Both models using supervised strategies brought better results than any other "no ground-truth" model at $\sigma = 0.05$. They reached a PSNR of $\approx 28.8$ dB.

The result obtained with vanilla EI didn't reproduce the result of [6]. We observed the difficulty of vanilla EI model to reconstruct the neuroimages at a noise level of 0.05 $\sigma$, despite using similar parameters than REI. This suggests that vanilla EI alone is not sufficient in this specific case. However, if vanilla EI didn't reach the expectations, the REI model achieved better results than what was expected. The obtained reconstructions with equivariant models might be sensitive to the image type, or to the two-channels problem that can bias or mislead the losses. This is discussed in 10. According to what was observed in this experiment, with a noise of 0.05 $\sigma$, REI has to be preferred than any other model that don't incorporate the ground-truth (TV, SSLEI).

It is worthy to mention that the fixed DnCNN model outperformed both equivariant models in different structure reconstructions.

### 8.3.2 Observations

Other tests were led to evaluate if using different parameters would lead to better results. Every statements are linked to observations made at $sigma = 0.05$ and might differ for other noise levels:

- Setting "no grad" to True resulted in faster convergence, but the learning sometimes demonstrated surprising behavior. After X epochs with a decreasing total loss, the total loss value went up suddenly (especially due to a jump of the EI loss, in favor of a smaller data fidelity). The model then failed to reach a better total loss than the previously reached minimum. This suggests the gradient reached an unstable area, where small variations in the learning rate (and stochastic variations) can cause damage to the training. Increasing the batch size, reducing the learning rate and using "ams grad" in the Adam optimizer reduced this effect.
- Setting "no grad" to False showed more stability than setting it to True. Especially, the EI loss was way more stable and kept decreasing, avoiding sudden jump. However, models using this parameter to False failed to accurately reconstruct the brains. After a certain point, the model hallucinates, with kind of "holes" appearing in the image (see Appendix) where coefficients were originally close to 0. This suggests that the solution found satisfied the loss minimization but not the expected visual reconstruction. After analysis, it appeared that some slices in the dataset contained very small areas with null coefficients and the model might have considered these low intensities as empty areas. This needs to be explored more as it doesn't explain why using no grad to True doesn't suffer from the same issue.

[1] *with pre-trained DnCNN

- Another ascertainment is the divergence between the loss optimum and the PSNR value with vanilla EI. In different experiments, the total loss reduced quickly but the PSNR failed to increase. It was even worse: with a lower loss, a lower PSNR. This suggest that the desired solution does not always converge with the function best solution. This is why REI must be preferred as it adds stability.
- Models using more transformations didn't show any gain in PSNR compared to 1 transformation only. As each image transformation in the EI loss is highly probably different from the others, we already scan a large range of transformations.
- A single attempt using a bigger DnCNN (depth to 12) in unrolled models didn't show an important gain. Same conclusion with more ADMM iterations (gain of 0.3 dB in supervised model).

### 8.3.3 Further Work

To complete the study and assess the robustness of equivariant imaging (EI and REI) with neuroimaging, we can reproduce a similar experiment but taking multiple random slices for each brain and not always the same one (at depth 0.52). For example, taking random slices between slice 48 and 208 to work on a wider variety of structures. This would require a bigger database to capture the different structures.

Also, the reason behing the "holes" appearing with the gradient in the EI loss is an issue we must reply to.

Finally, it would be worthy to apply the same models with more data and different noise levels to assess the quality of SSLEI models with MRI neuroimaging.

# Chapter 9

# Dynamic Models

The main purpose of this study is to assess the relevance of Dynamic MRI reconstruction with equivariant imaging, implying the challenging task to upgrade to 3D. It is then needed to understand how to simulate Dynamic MRI. This type of scan is usually acquired over a single same frame (or volume) over time in clinic. As we do not possess such a database, we simulate a kind of Dynamic MRI by flattening 3D volumes as a sequence of images. It consists in using the depth (here time 'T') dimension as the batch size of a volume. We then apply a time-varying mask over the flattened batch of images. From a 256x256x256 volume we now have for a single volume a tensor of shape (B*T,C,H,W) = (256,2,256,256). With the reverse operation we can rebuild the volume. A ground-truth volume, displayed every 4 slices:

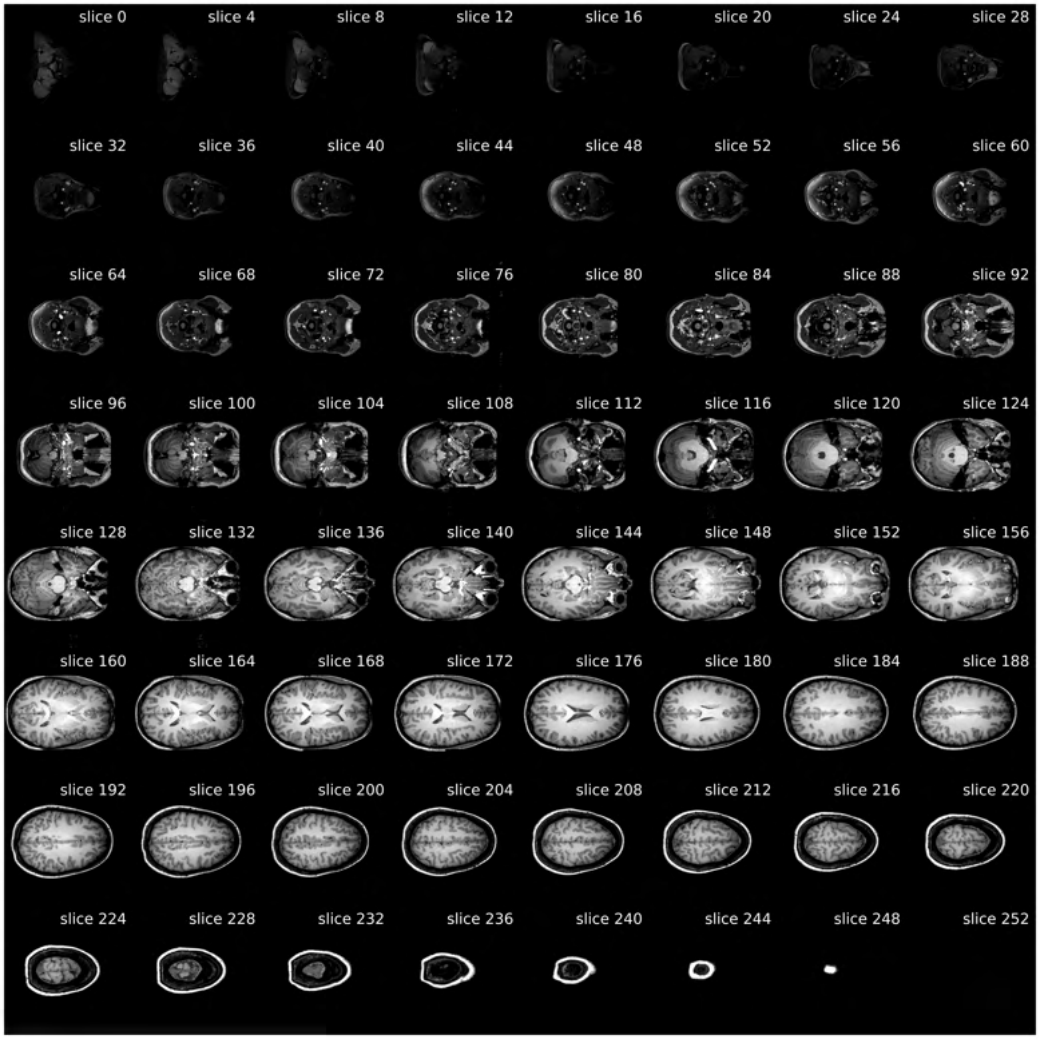



Figure 9.1: Flattened Brain

As we don't have relevant features over the firsts and lasts frames of most participants brains, we only work on frames from 48 to 208 (128+/-80). This skips the empty sequence slices as well as the few features frames, while still working on various types of head slice. Applying a time-varying MRI Cartesian acceleration mask using Gaussian undersampling ($R = 4$ and $\sigma = 0.05$):

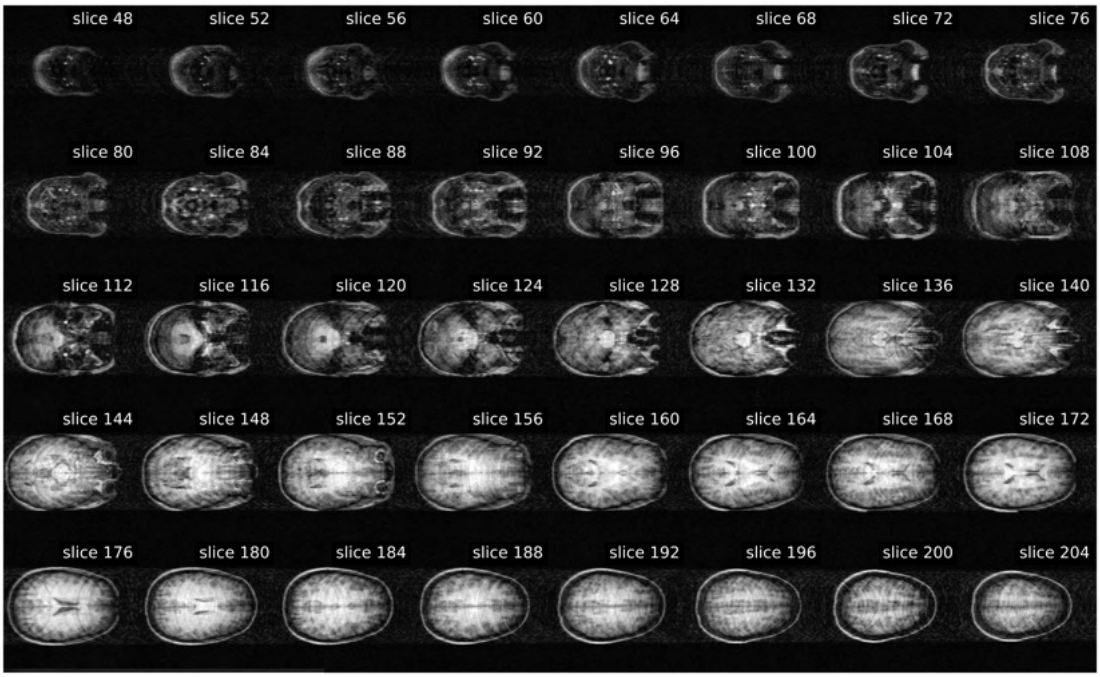



Figure 9.2: Flattened Sub-sampled Brain Linear Reconstruction

It is not because the volumes are flattened into a batch of 2D images to degrade them in the k-space that we apply 2D denoising architectures. Dynamic MRI implies capturing a sequence of 2D image slices over time, therefore over a 3D volume. The additional dimension means we now work on voxels and not pixels, carrying by definition more information. 3D denoisers can consequently better model and reduce noise by utilizing the additional information contained in this third dimension, leading to (an expected) improved image quality and more accurate reconstructions. Especially, consecutive frames often contain redundant information helping to smooth the sequence from one frame to another. We expect 3D denoisers to better leverage this redundancy to preserve dynamic structures and motion patterns in the MRI data, unlike 2D denoisers which treat each frame independently. The key advantage of 3D denoisers is their ability to utilize the full 3D structure to improve denoising and reconstruction.
The 3D denoisers do not make any distinction between a 3 spatial dimension volume or a 2D + time volume, this is why the use of 3D denoisers does not alter the physical processes underlying MRI and respects our initial objective. We refer to depth of 3D denoiser (H,W,D) but the time of a dynamic sequence (H,W,T).
The denoised sequences files (videos) are available in attached documents. Models are evaluated over 20 volumes, using the lowest loss checkpoint for unrolled models. We do not change the parameters of fixed models depending on the noise level. We only present visual results for $\sigma = 0.05$ but other noise levels are in table for each section. As a reference, linear reconstructions are:

| $\sigma$ | $A^\dagger$ |
|---|---|
| 0.0 | 23.6±0.4 |
| 0.01 | 23.6±0.4 |
| 0.05 | 22.7±0.3 |
| 0.1 | 20.7±0.2 |

Table 9.1: 3D Linear Reconstruction Noise Comparison

## 9.1 3D Fixed Parameters Models

### 9.1.1 3D TV-ADMM

The current Deep Inv library only contains a 2D TV Denoiser. Extending the DnCNN code to 3D, considering the depth, we can apply a similar TV-ADMM than in 8.1.1 with fixed parameters for every noise level:

Table 9.2: Parameters for 3D TV-ADMM

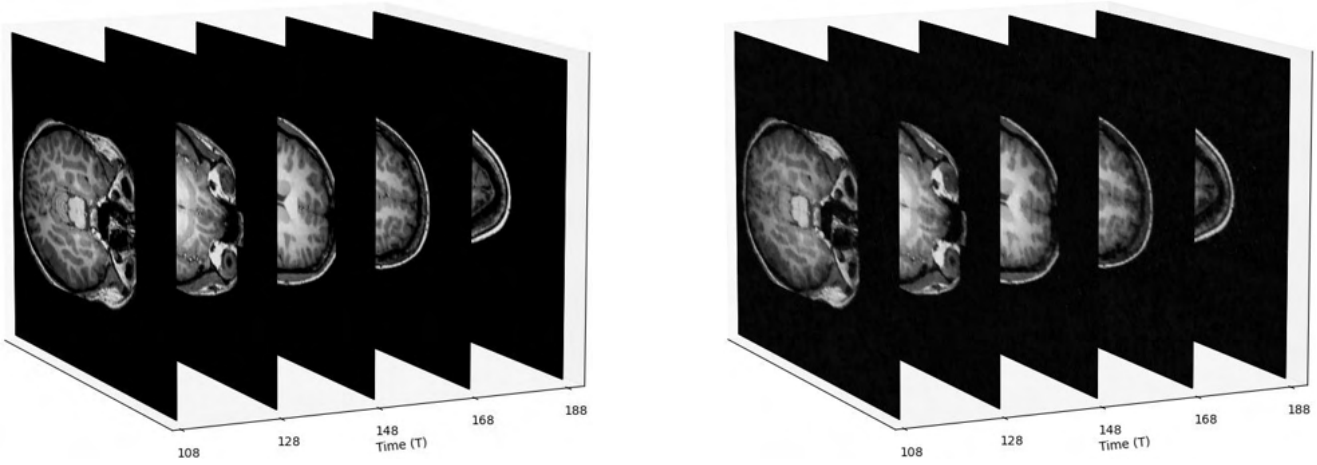
| **Parameter** | **Value** |
|---|---|
| TV Prior Maximum Iterations | 40 |
| Step Size | 0.5 |
| Regularization Parameter ($\lambda$) | 0.015 |
| Maximum Iterations | 30 |

We do not try to find the best parameters for each noise level due to the fastidious and long task. Keeping the same parameters results for different $\sigma$ in :

| $\sigma$ | $A^\dagger$ | TV-ADMM |
|---|---|---|
| 0.0 | 23.6±0.4 | 26.4±0.2 |
| 0.01 | 23.6±0.4 | 26.4±0.2 |
| 0.05 | 22.7±0.3 | 26.2±0.3 |
| 0.1 | 20.7±0.2 | 24.5±0.2 |

Table 9.3: 3D TV-ADMM Noise Comparison

We recall that a 3 dB decrease in PSNR means that the noise power (as measured by MSE) has doubled (explained in 7.4.2).
The quality of the linear reconstruction decreases by more than 3 dB between the noise-free (but 4x accelerated) and 0.1 $\sigma$ volumes, but decreases of only 2 dB with the denoised volumes. This suggests that if the noise hads doubled between $\sigma = 0.0$ and $\sigma = 0.1$, the TV-ADMM denoising mitigated this effect by only $\approx 1.5$.



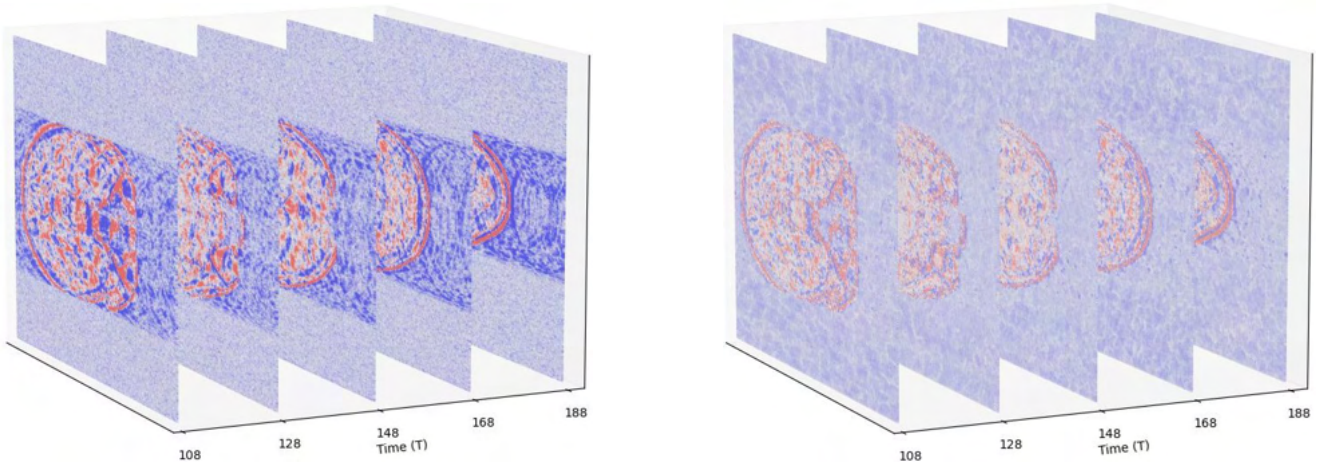

(a) Original volume

(b) Volume after denoising

Figure 9.3: 3D TV-ADMM Reconstruction

Visually, comments are similar to the ones for 2D TV-ADMM 8.1.1. Edges are somewhat blurred and the acceleration artifacts are only partially corrected. The error ($error = ground\text{-}truth - reconstruction$) is normalized from $[-0.5, 0.5]$ to the range $[0, 1]$, and values beyond $\pm 0.5$ are clipped to saturate the color

map. The intensity and hue of the color map provide insights into how well the reconstruction model performs, with blue suggesting an overestimation and red indicating an underestimation of the ground truth values. Here the error around the brain looks like spots with a too strong correction.




(a) Linear Error

(b) Reconstruction Error

Figure 9.4: Error in 3D TV-ADMM Model

### 9.1.2 3D PnP-ADMM with 3D DnCNN

Aiming to use a DnCNN, to extend the code to 3D considering the depth, we need to replace the conv2D layers by conv3D layers. We do not use the batch normalisation as it tends to overfit in unrolled architectures (observed in this specific study, might not be a general statement).
Working on a batch of two-channels 256x256x160 volumes is too expensive for most computers. Training a DnCNN is then impossible without patching. We can re-use the same strategy as in 8.1.2 consisting of spatial patches after subsampling the k-space. We degrade the volumes with the physics and extract 3D patches by extending the current "patch extractor" function. The new addition accepts a patch size in the tuple format, allowing patches with different height, width and depth. Using this 3rd dimension multiplies the possible number of patches suggesting we better set a maximal number of patches. For each volume we use a fixed number of patches following $H^3/patchsize^3$. Having already trimmed the depth is not an issue.

Regarding the similar results obtained with Gaussian noise and Rician noise in 8.1.2, we only work with a DnCNN learning a mapping for Gaussian noise in the image space (easier to implement). We use only 20 patched volumes to train the DnCNN, and test it over 10 patched volumes. We first used 64x64x64 patches but the results were way worse than using 32x32x32 patches (difference of $\approx 4$ dB at $\sigma = 0.05$). We adjust the DnCNN parameters by reducing its depth (number of layers). Batch normalization is still disabled for overfitting issues. We only train over 5 epochs:

Table 9.4: Parameters for 3D DnCNN

| **DnCNN config** | **Value** |
|---|---|
| Batch Normalization | False |
| Depth | 7 |
| Number of Filters (nf) | 64 |
| **Training Parameters** | |
| Epochs | 5 |
| Learning Rate | 5e-4 |

Presented values are computed over the patches and not the complete volumes, explaining the high variance. It's not worthy to try to learn a DnCNN with noise-free volumes:

| $\sigma$ | $A^\dagger$ Patches | DnCNN Patches |
|---|---|---|
| 0.01 | 38.1±1.2 | 43.6±3.2 |
| 0.05 | 24.1±1.2 | 35.9±4.9 |
| 0.1 | 20.0±0.0 | 30.2±3.3 |

Table 9.5: 3D TV-ADMM Noise Comparison

Higher noise models would have benefited from more epochs. We observe a quick PSNR drop as the noise is increasing, with patches $\approx 16$ times more noisy at $\sigma = 0.1$ than at $\sigma = 0.01$.
To test the PnP-ADMM model with the pre-trained 3D DnCNN, we use the same parameters as for 2D PnP-ADMM but with less iterations:

Table 9.6: Parameters for fixed 3D PnP-ADMM

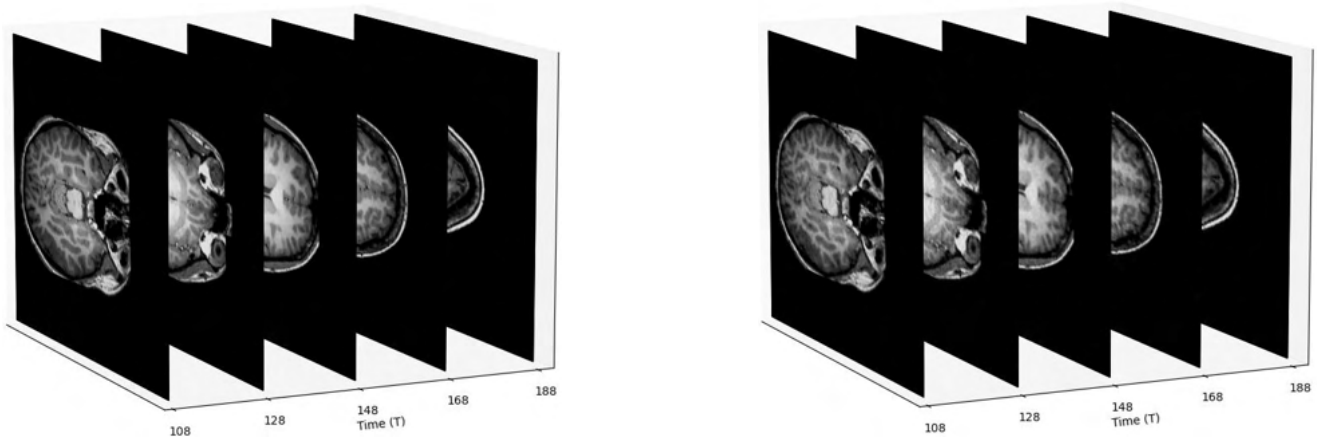
| **ADMM Parameters** | **Value** |
|---|---|
| stepsize | 1.75 |
| beta ($\beta$) | 1.25 |
| max iter | 20 |

When applying an optimization procedure to the volumes at a specific noise level, we use the pre-trained DnCNN at this noise level. It might not be the best option and require more rigorous researches. We didn't go deeper to save time to complete every tasks and focus on SSLEI models.

| $\sigma$ | $A^\dagger$ | Fixed 3D DnCNN PnP-ADMM |
|---|---|---|
| 0.01 | 23.6±0.4 | 29.7±0.7 |
| 0.05 | 22.7±0.3 | 30.7±0.5 |
| 0.1 | 20.7±0.2 | 27.8±0.4 |

Table 9.7: 3D PnP-ADMM Noise Comparison

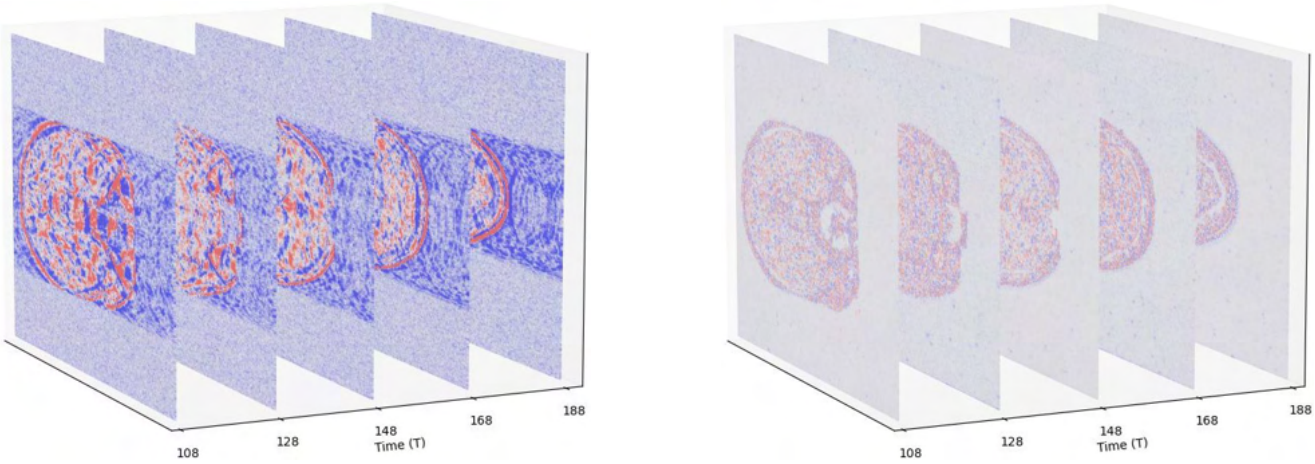
The selected parameters were optimized for 2D PnP-ADMM at noise level 0.05. This might explain why results are not convincing at every noise levels. As it was not the purpose of this study, we did not look for better parameters at different noise levels.




(a) Original volume (b) Volume after denoising

Figure 9.5: 3D fixed DnCNN PnP-ADMM Reconstruction

Visually, we obtain a very convincing reconstruction when parameters are adjusted to the specific noise level. The edges are sharp and acceleration artifacts are entirely corrected, but with diffuse noise in high intensity areas and persistent "constellation" noise around the brain.




(a) Linear Error

(b) Reconstruction Error

Figure 9.6: Error in 3D fixed DnCNN PnP-ADMM Model

## 9.2 3D Unrolled Models

3D unrolled models in full-resolution are the most consuming models we work on in this study. It is well known that unrolling is challenging for GPU usage. Similarly to unrolled 2D models, we can't easily use patches in the Trainer but we have no other choice or we saturate the GPU memory too fast. Then, the easiest way to avoid saturating the memory is to pass patches in the dataloader and then reconstruct the whole volume after putting the patches in the trained model. But what patches?

We can't extract patches in the Fourier space because in the frequency domain, the information about the exact spatial location of features is lost. However, Fourier is time-independent (Fourier isn't related to time but only to space). As we work on Dynamic MRI, we use a 3rd temporal dimension which is not considered as spatial. This is over this axis that we can patch the data to limit memory usage.

These patches depth "$t$" can be adjust to fit the GPU capacity, but we consider that under $t = 8$ slices in depth, it would not have enough information to capture the dynamic aspect.
In this experiment, we set the patch depth to $t = 8$, meaning we work on smaller volumes of shape 256x256x8 instead of 256x256x160. Each volume then have $160/8 = 20$ patches. This suggest that with an initial batch of B volumes, we go from tensors of shape $(B, 2, T, H, W)$ to $(B\dot{(}T/t), 2, t, H, W)$. Here, for reproducibility: from $(B, 2, 160, 256, 256)$ to $(B\dot{(}160/8), 2, 8, 256, 256)$.

We need to be careful with the acceleration mask we use so each patched mask $T/t$ is different inside the batch of patches, but is repeated with a modulo $T$. We still generate a mask with the depth of a single volume, so we apply the physics on the volumes before patching. We patch both measurements and masks to pass them in a dataloader, alongside the ground-truth for eval. Using these temporal patches, we recommend setting the train and test batch size to 1. We can interpret the patch as a replacement of the batch size. If you can set train batch size = 2, we recommend increasing $t$. In ideal scenario, $t = T$.
In all the experiments we used a $t$ that can divide $T$.
The number of patches per volume has to be kept in memory in order to rebuild them afterward, and compute the PSNR.

Table 9.8: Parameters for 3D unrolled PnP-ADMM

| **DnCNN config** | **Value** |
|---|---|
| Batch Normalization | False |
| Bias | True |
| Depth | 7 |
| Number of Filters (nf) | 32 |
| **Iterator Parameters** | |
| iterator | ADMM |
| iter | 5 |
| **Training Parameters** | |
| Epochs | $< 65$ |
| Learning Rate | 5e-4 |

Parameters of ADMM are lists of length "iterator iter" so here 5, and initialized to 1, except for sigma denoiser to 0.1.

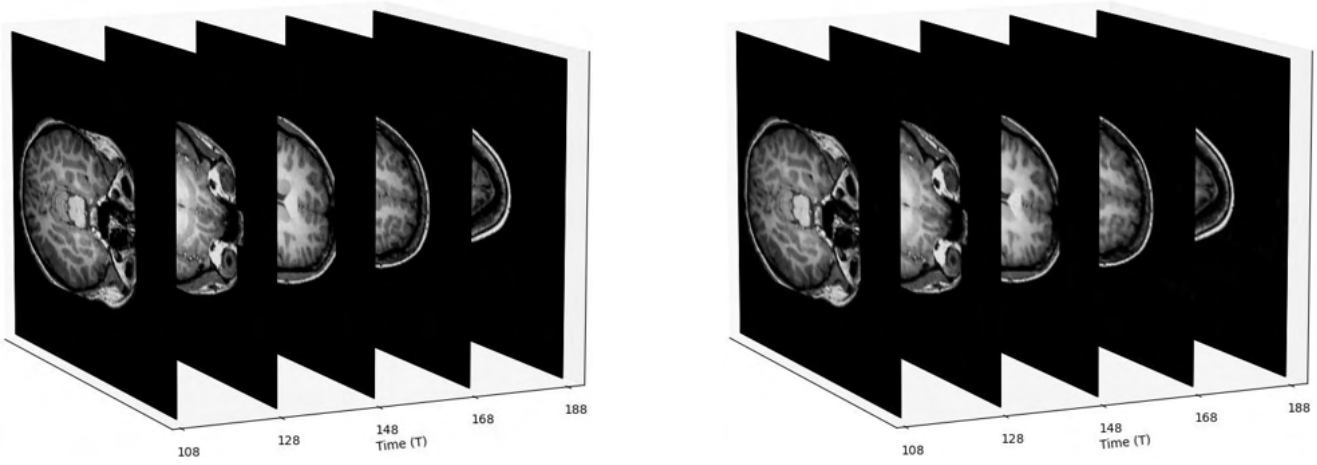
### 9.2.1 3D unrolled supervised PnP-ADMM

Supervised models has been trained for a maximum of 30 epochs. Presented results are computed for minimum loss over the 30 epochs.

| $\sigma$ | $A^\dagger$ | Supervised 3D |
|---|---|---|
| 0.0 | 23.6±0.4 | 34.9±0.6 |
| 0.01 | 23.6±0.4 | 33.9±0.5 |
| 0.05 | 22.7±0.3 | 30.8±0.4 |
| 0.1 | 20.7±0.2 | 29.3±0.4 |

Table 9.9: 3D Supervised Noise Comparison

The results are exceptional for every noise level. It significantly outperforms the pre-trained DnCNN models for noise different than 0.05. Remarkably, the complex PSNRs from the supervised models are consistently about 10 dB higher than those achieved with backprojection. The proposed model is then very efficient for this task and it appears to be challenging for self-supervised models to reach such high values.

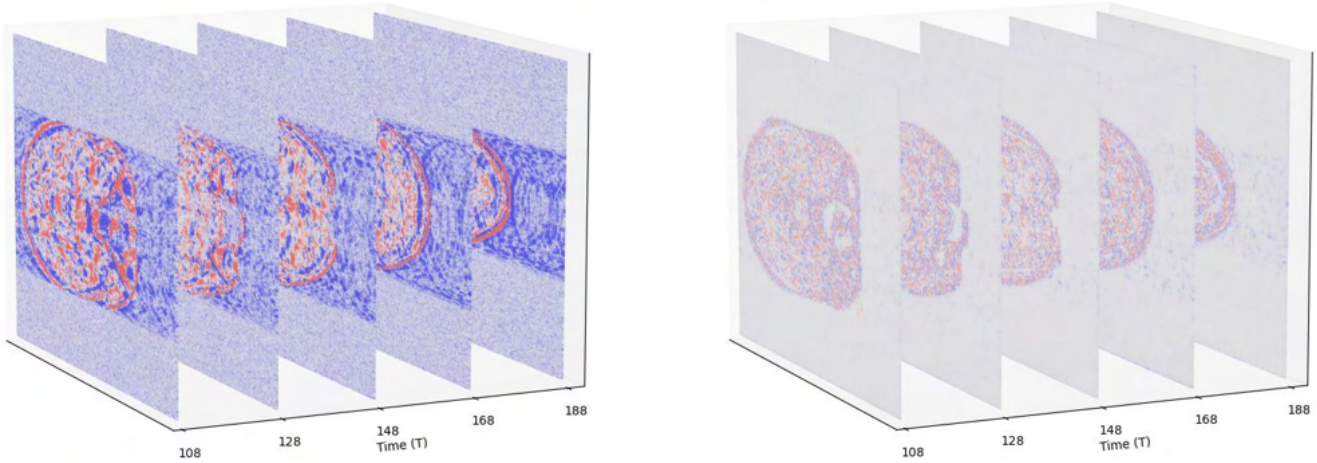



(a) Original volume

(b) Volume after denoising

Figure 9.7: 3D Supervised Reconstruction

The model is even able to correct the original acquisition artifacts! The noise inside the brain is substantial, and spots of parasite noise persists over the acceleration artifacts outside the brain. The better noise handling from the fixed DnCNN model in the acceleration artifacts can be explained by the pre-trained DnCNN especially for the denoising task. Also, the PnP-ADMM model with the fixed DnCNN applies more ADMM iterations.




(a) Linear Error

(b) Reconstruction Error

Figure 9.8: Error in 3D fixed Supervised Model

### 9.2.2 3D unrolled unsupervised PnP-ADMM EI

From this point we explore models that have for sure never been implemented before, and every comments and conclusions will be discussed in later sections. We don't have any reference to compare our results with past experiments results. We distinguish the no time flip EI version (transforms only in spatial domain) from the dynamic EI (with flip) version. We define the dynamic EI in this paper as "EI Flip". The two sub-types are compared to assess the relevance of the time flip. Also, models were trained only once for every noise level, and the lowest loss checkpoint was kept for evaluation. More training are required to improve the accuracy of the proposed results. Decision was taken to use "no grad" = True due to 2D results. For both 3D vanilla EI models we use the parameters:

Table 9.10: Parameters for 3D EI loss

| **EI Loss Parameters** | |
|---|---|
| Transform | Affine |
| Number of transformations | 1 |
| No Grad | True |
| Apply Noise | False |
| Weight EI loss ($\alpha$) | 1.0 |

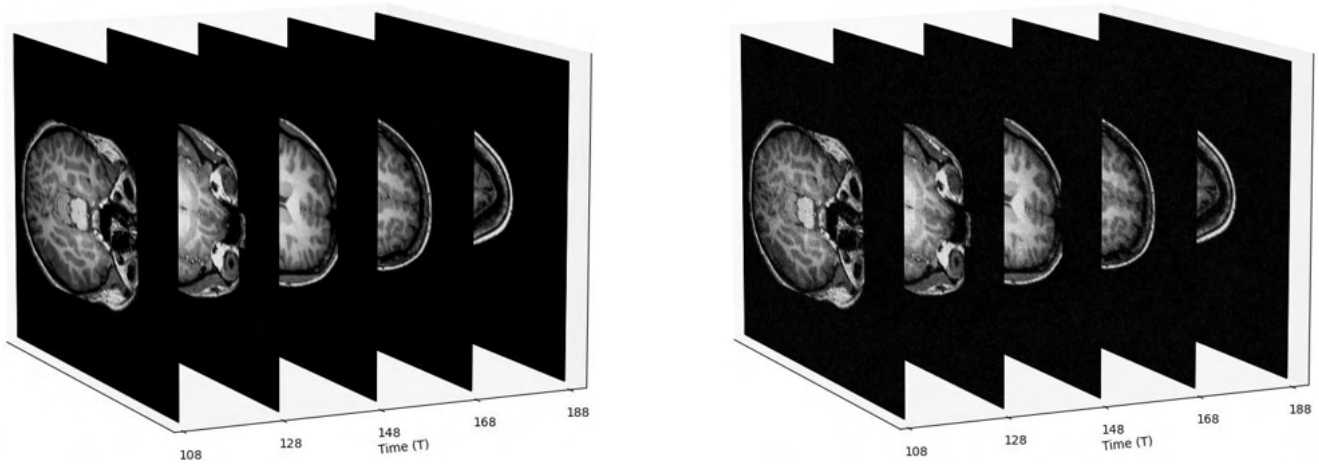
#### No Flip

This version only includes the spatial affine transformations exposed previously and do not use any temporal symmetry. Therefore we have no transformation in the temporal domain. In the EI loss, each small patch of dimensions 256x256x8 is transformed according to a random affine transformation and this same transformation is applied on every slice of the small batch. This suggests that the transformations between two successive patches are different, and we have in total $T/t \times B = 2000$ different transformations applied over each epoch ($160/8 \times 400$).
Regarding the no flip version, we observe the same noise sensitivity of the model with the PSNR decreasing as the noise is increasing. However, the PSNR values and drop is different. At low noise level,

| $\sigma$ | $A^\dagger$ | EI |
|---|---|---|
| 0.0 | 23.6±0.4 | 32.2±0.6 |
| 0.01 | 23.6±0.4 | 31.5±0.6 |
| 0.05 | 22.7±0.3 | 26.5±0.2 |
| 0.1 | 20.7±0.2 | 21.7±0.1 |

Table 9.11: 3D Unsupervised EI no flip Noise Comparison

EI no flip is very good with a gain close to 10 dB and relatively close to the supervised model, but the model experiments a very big drop of dB at 0.1 $\sigma$ with a PSNR just 1 dB over the backprojection.

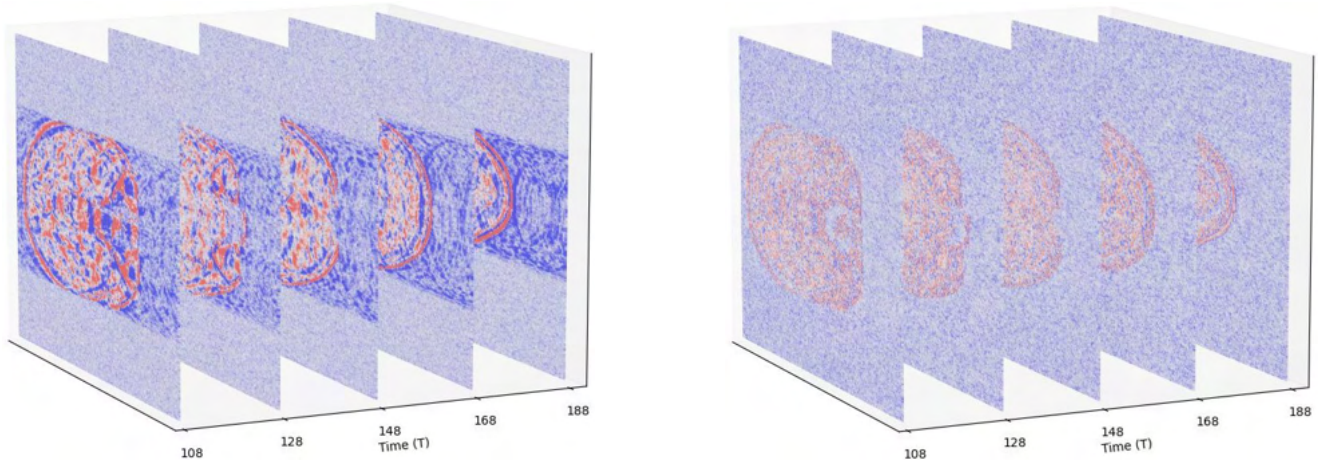



(a) Original volume

(b) Volume after denoising

Figure 9.9: 3D EI no flip Reconstruction

The obtained video at $\sigma = 0.05$ is still very convincing! The sequence is exploitable but suffers from small noise perturbations over the whole volume. They nevertheless do not alter too much the visualisation. The acceleration artifacts are already very well corrected according to the error difference.




(a) Linear Error

(b) Reconstruction Error

Figure 9.10: Error in 3D EI no flip Model

**Flip**

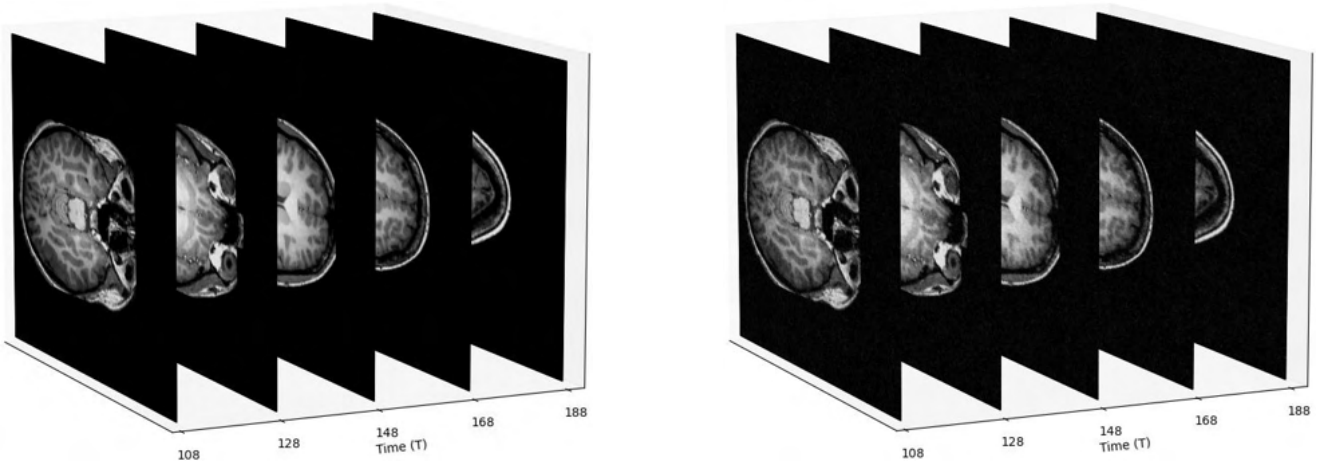
The flip version differs from the previous one by implementing the newly introduced "time flip" (or dynamic EI) over the sequences. In addition to the previous transformation, an independent time flip

can be applied over each small batch. We use a Bernoulli drawing with a probability of 50% for each small batch to apply the flip. This suggests that we expect to have $\approx X = 2000/2 = 1000$ flipped patches in one epoch. We recall that this transformation is orthogonal to the spatial plans and therefore do not alter the spatial transformations. We obtain:

| $\sigma$ | $A^{\dagger}$ | EI Flip |
|---|---|---|
| 0.0 | 23.6±0.4 | 33.1±0.6 |
| 0.01 | 23.6±0.4 | 32.0±0.5 |
| 0.05 | 22.7±0.3 | 26.6±0.2 |
| 0.1 | 20.7±0.2 | 21.9±0.1 |

Table 9.12: 3D Unsupervised EI Flip Noise Comparison

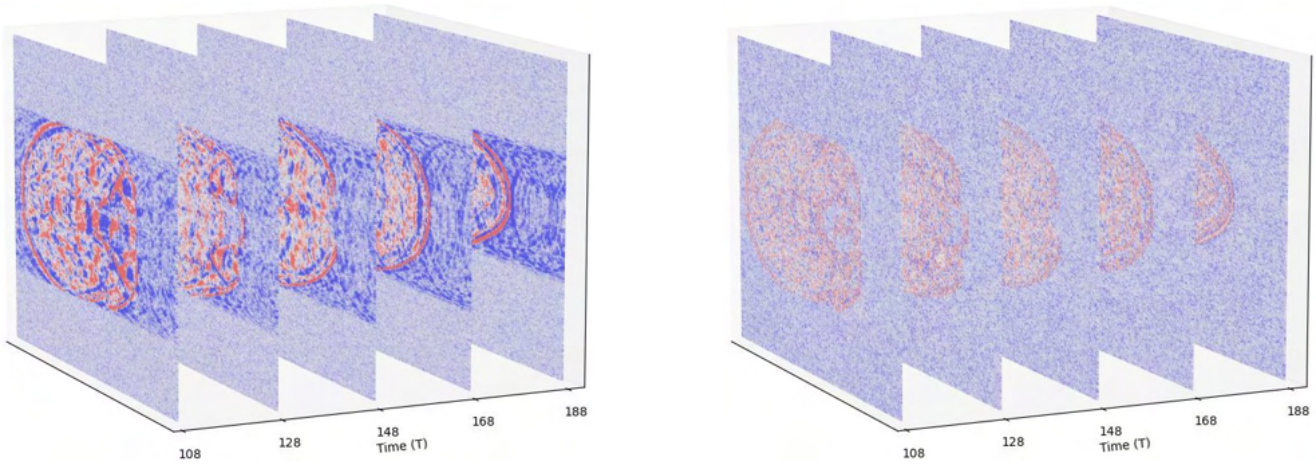
Results are somewhat similar to no-flip model but still we observe a non-negligible gain at $\sigma = 0$ and $\sigma = 0.01$ with respective improvements of 0.9 dB and 0.5 dB. The higher the noise is, the less relevant is the effect.




(a) Original volume (b) Volume after denoising

Figure 9.11: 3D EI Flip Reconstruction

Visually, the small PSNR gain is not human-eye visible and results in sequences extremely similar to the no-flip version. The error at 0.05 $\sigma$ appears to be the same. Just the noise distribution is different.




(a) Linear Error (b) Reconstruction Error

Figure 9.12: Error in 3D EI Flip Model

**Comparison**

The dynamic vanilla EI models demonstrated better reconstruction results than the 2D models. This difference can be mainly explained by the intrinsic properties of using volumes and not just slices. There is a smoothing operated from one slice to another and the information is correctly exploited by the model.

| $\sigma$ | $A^\dagger$ | EI | EI Flip |
|---|---|---|---|
| 0.0 | 23.6±0.4 | 32.2±0.6 | 33.1±0.6 |
| 0.01 | 23.6±0.4 | 31.5±0.6 | 32.0±0.5 |
| 0.05 | 22.7±0.3 | 26.5±0.2 | 26.6±0.2 |
| 0.1 | 20.7±0.2 | 21.7±0.1 | 21.9±0.1 |

Table 9.13: 3D Unsupervised EI Noise Comparison

Regarding the relevance of the flip in the temporal domain, results are mitigated. If there is an obvious gain at small noise levels, this gain disappears as the noise increases. However it is now demonstrated that **incorporating this transform is preferable in vanilla EI** as it can only improve the reconstruction. Especially at low noise level, the EI loss was able to exploit this new transformation into knowledge for reconstruction.

### 9.2.3 3D unrolled unsupervised PnP-ADMM REI

The REI models only differ from the EI models by the use of the SURE loss to replace the MC loss. The EI loss part remains the same, at the exception of using the "apply noise" = True instead of False, to be in the context of robust equivariant imaging as defined by [6]. Therefore parameters are:

Table 9.14: Parameters for 3D REI loss

| **EI Loss Parameters** | |
|---|---|
| Transform | Affine |
| Number of transformations | 1 |
| No Grad | True |
| Apply Noise | True |
| Weight EI loss ($\alpha$) | 1.0 |

In following results, we use the epoch with the lowest loss obtained over the 65 trained epochs. It might not be the last epoch. Also, at noise level $\sigma = 0$, we use the results of EI as REI = EI without noise by definition. If differences are obtained, they would only be due to the training stochastic aspect (random samples).

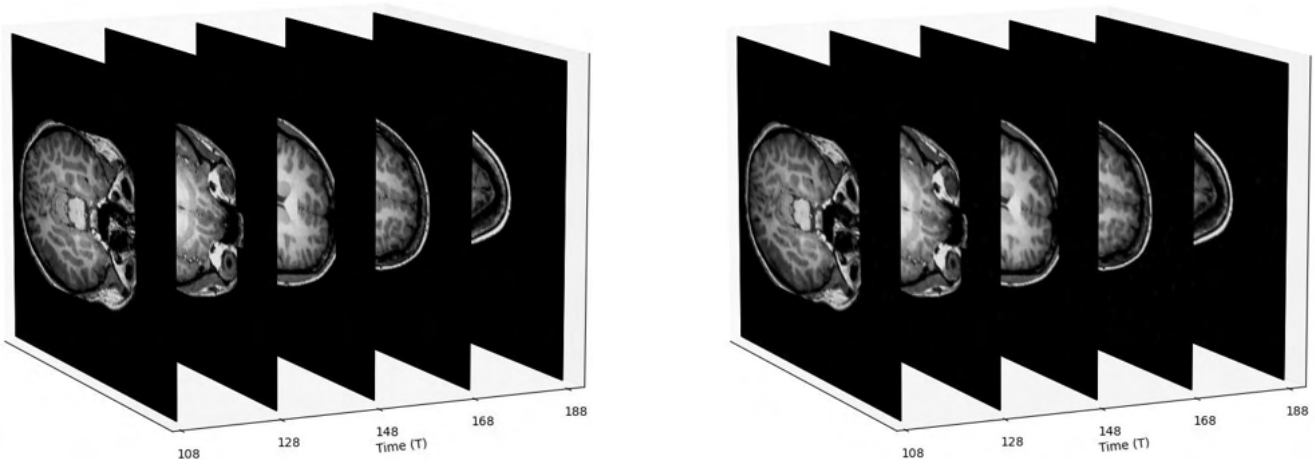
If the trend of REI in dynamical models is the same as for 2D (in paper [6]), we should observe a gain as soon as we have noise and especially for strong noise. In our own experiment in 2D 8.3, we had a gain of 4 dB at $\sigma = 0.05$ for REI. We were also very close to the supervised unrolled model.

**No Flip**

| $\sigma$ | $A^\dagger$ | REI |
|---|---|---|
| 0.0 | 23.6±0.4 | 32.2±0.6 |
| 0.01 | 23.6±0.4 | 30.6±0.5 |
| 0.05 | 22.7±0.3 | 29.8±0.5 |
| 0.1 | 20.7±0.2 | 28.4±0.4 |

Table 9.15: 3D Unsupervised REI no flip Noise Comparison

We already obtain impressive results at high noise levels! We are just one decibel under the supervised model at $\sigma = 0.05$ and 0.1. However, we have a very surprising value at $\sigma = 0.01$, with a result which is 1 dB worse than the EI model without flip.

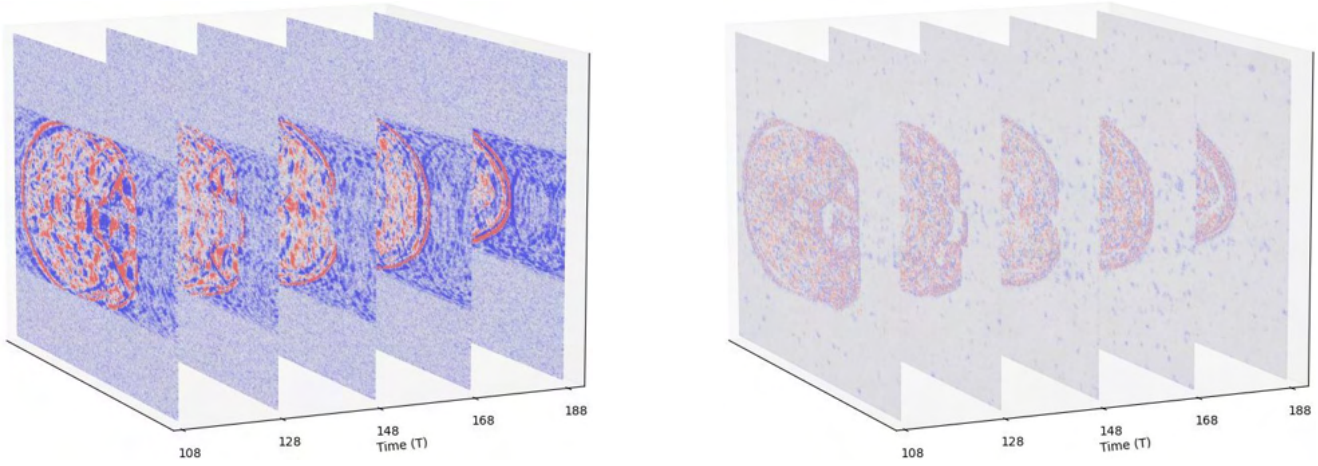



(a) Original volume

(b) Volume after denoising

Figure 9.13: 3D REI no flip Reconstruction

The visual reconstruction is more that convincing with video sequences entirely denoised in and around the brain. The complex structures inside the brain are also accurately reconstructed, and no acceleration artifact is present. At sigma $\sigma = 0.05$, differences between the supervised model and REI no flip are not easy to catch. The error shows a very good denoising with residual noise in the central band (i.e. containing acceleration artifacts). The main visible difference with EI is the denoising around the brain.




(a) Linear Error

(b) Reconstruction Error

Figure 9.14: Error in 3D REI no flip Model

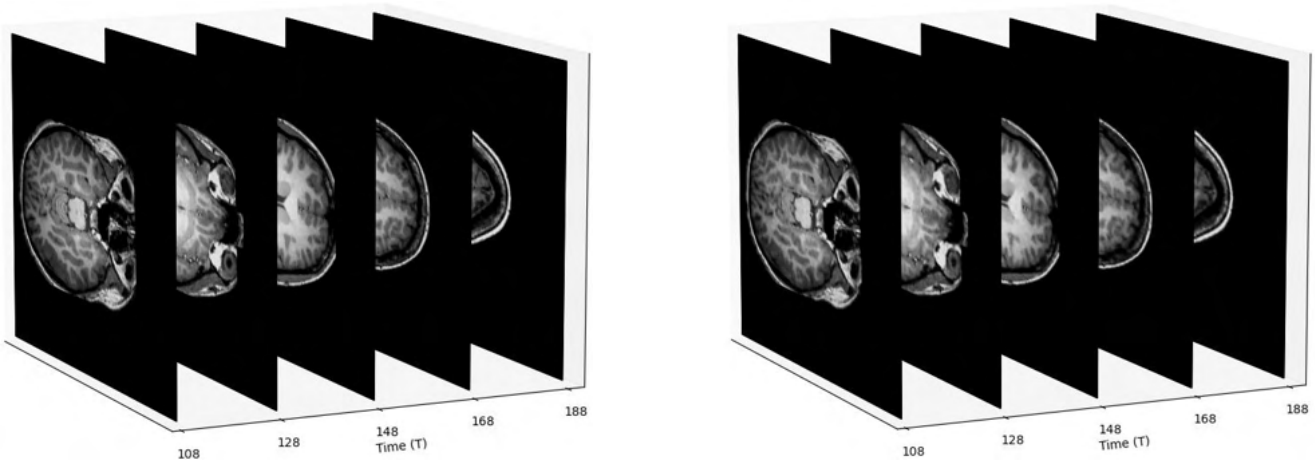
### Flip

The flip version integrates the dynamic transform in the EI loss with the same logic as exposed in 9.2.2.

| $\sigma$ | $A^\dagger$ | REI Flip |
|---|---|---|
| 0.0 | 23.6±0.4 | 33.1±0.6 |
| 0.01 | 23.6±0.4 | 31.1±0.5 |
| 0.05 | 22.7±0.3 | 29.8±0.4 |
| 0.1 | 20.7±0.2 | 28.6±0.4 |

Table 9.16: 3D Unsupervised REI Flip Noise Comparison

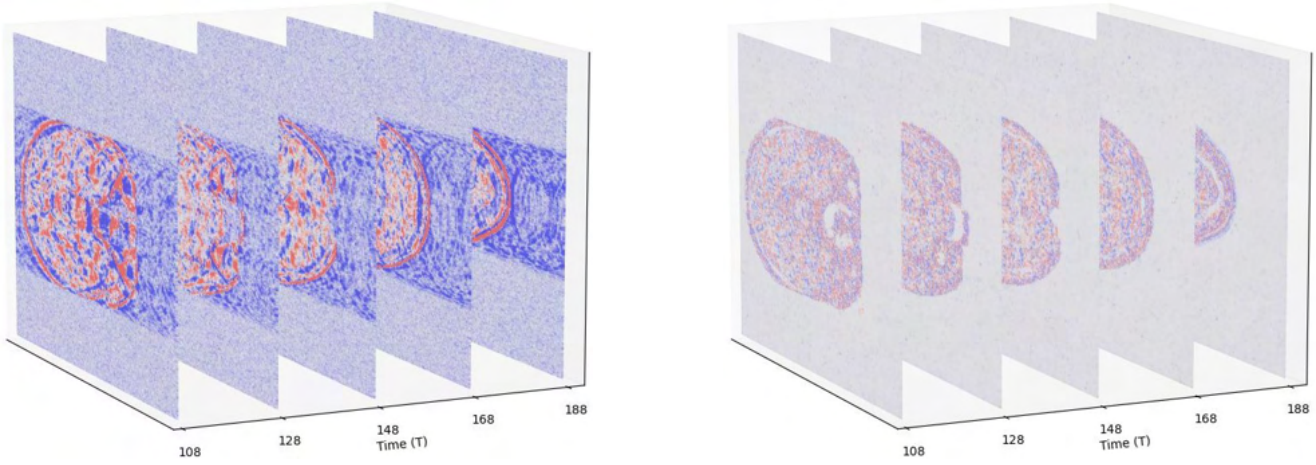
Obtained results are once again better (at least never worse) than the REI no flip version but do not show an amazing improvement. But still, there is some. At noise level 0.01, the disappointed PSNR for REI no flip is the best improvement with an increase of 0.5 dB. The gain from no flip version disappears once again as the noise increases.




(a) Original volume (b) Volume after denoising

Figure 9.15: 3D REI Flip Reconstruction

The sequences obtained are comparably the same as for no flip version, but this time it's entirely denoised around the brain structure (at least until 0.05 $\sigma$). There is no residual noise and just some coefficients are badly estimated by the model but it's very unsignificant and the model is very satisfying.




(a) Linear Error (b) Reconstruction Error

Figure 9.16: Error in 3D REI Flip Model

**Comparison**

| $\sigma$ | $A^\dagger$ | REI | REI Flip |
|---|---|---|---|
| 0.0 | 23.6±0.4 | 32.2±0.6 | 33.1±0.6 |
| 0.01 | 23.6±0.4 | 30.6±0.5 | 31.1±0.5 |
| 0.05 | 22.7±0.3 | 29.8±0.5 | 29.8±0.4 |
| 0.1 | 20.7±0.2 | 28.4±0.4 | 28.6±0.4 |

Table 9.17: 3D Unsupervised REI Noise Comparison

However, we can once again state that **incorporating the time transform is preferable in REI** and can only provide gain in the reconstruction. Possible reasons for the low gain as the noise increases are given in later sections. If the gain from EI to REI is not that big, the difference between linear reconstructions and REI models are impressive with always more than 7 dB of difference, suggesting we have 5 times less noise (7.4.2)! Keeping in mind this model doesn't have access to the ground-truth data, the results are outstanding and flirt with what we obtain in supervised models.

## 9.3 3D Comparison and further work

### 9.3.1 Comparison

| $\sigma$ | $A^\dagger$ | TV-ADMM | PnP-ADMM* | EI | EI Flip | REI | REI Flip | Supervised |
|---|---|---|---|---|---|---|---|---|
| 0.0 | 23.6±0.4 | 26.4±0.2 | N/A | 32.2±0.6 | 33.1±0.6 | 32.2±0.6 | 33.1±0.6 | 34.9±0.6 |
| 0.01 | 23.6±0.4 | 26.4±0.2 | 29.7±0.7 | 31.5±0.6 | 32.0±0.5 | 30.6±0.5 | 31.1±0.5 | 33.9±0.5 |
| 0.05 | 22.7±0.3 | 26.2±0.3 | 30.7±0.5 | 26.5±0.2 | 25.7±0.2 | 29.8±0.5 | 29.8±0.4 | 30.8±0.4 |
| 0.1 | 20.7±0.2 | 24.5±0.2 | 27.8±0.4 | 21.7±0.1 | 21.9±0.1 | 28.4±0.4 | 28.6±0.4 | 29.3±0.4 |

Table 9.18: PSNR Comparison of 3D models

If the supervised models are the best ones at reconstruction (when adjusted parameters are provided for fixed parameters models), the REI models appear to be very good candidates with less than two decibels of difference with supervised models (except for $\sigma = 0.01$). Notably, as noise levels increase, the REI models perform increasingly closer to the supervised models compared to their performance at lower noise levels. TV still appears to be a good alternative to vanilla EI when the noise level increases, but can't compete with REI. With adjusted parameters it might still be better than vanilla EI for every noise level! The pre-trained DnCNN models can also surely outperform REI when the parameters are accurately defined. However, it might be long to find the perfect adjustments and unrolled supervised model can be preferred if ground-truth data is available.

Also, as a general statement, **incorporating the time transform is preferable in dynamic self-supervised learning with equivariant imaging**. Even if we do not have a big gain at high noise level, it always showed improvements, either in EI or REI.

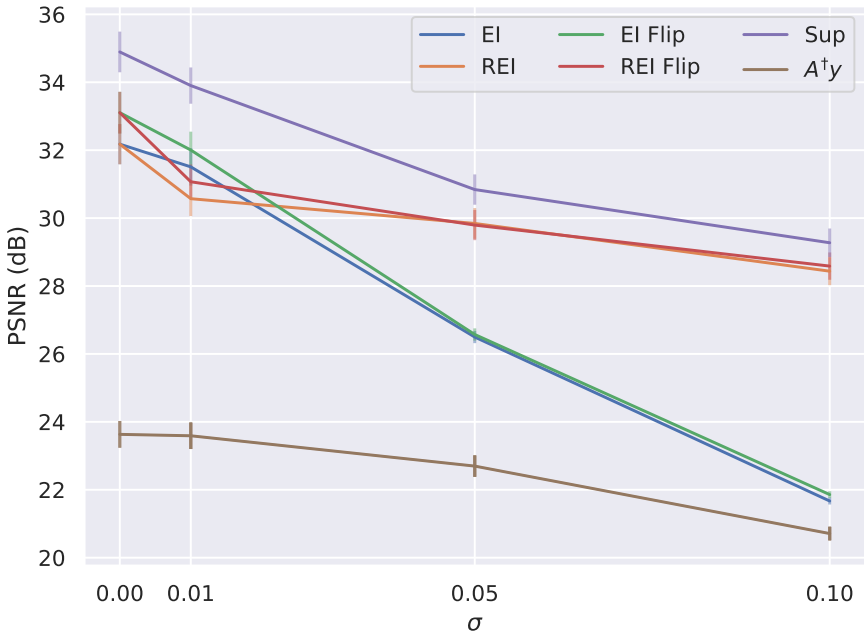



Figure 9.17: Unrolled Models Comparison for different $\sigma$

Focusing on unrolled models only, trends are clearer when computing the different curves by noise level. It also helps to identify behaviors that seem to be unexpected and also the further work to accomplish.

EI vs. REI: The difference between EI and REI models becomes more pronounced with higher noise levels. EI struggles significantly with noise levels above $\sigma = 0.01$, while REI models continue to perform well even under high noise conditions, demonstrating its robustness. However, EI is surprisingly better at low noise level than REI.

Flip vs. No Flip: For vanilla EI models, the flip transformation provides a substantial improvement at all noise levels. In particular, there's a 1 dB gain in the noise-free model, indicating that temporal transformations effectively contribute to self-supervised learning. For REI models, the impact of the flip is less pronounced. At low noise levels, the benefit of the flip is relatively significant, but as noise increases, the difference becomes less noticeable. This reduced impact can be attributed to the robust nature of the REI model, which, due to its use of the SURE loss for consistency and noise application in the EI loss, diminishes the importance of temporal transformations at higher noise levels.

### 9.3.2 Answers to initial questions

We can now reply to the questions stated in the introduction.

*Is dynamic EI able to capture very fine structures in brain MRI?*
Reconstructions showed impressive results and this for every noise level. Particularly at high noise levels, REI incorporating the dynamic transform demonstrated the relevance of equivariant imaging to compete with supervised models which is a massive achievement. Therefore the answer is yes.

*How does the EI loss deal with a new dimension?*
This question is more nuanced. We obtained results that were not very stable due the absence of gradient in the EI loss in the conditions of our experiment. Therefore if the EI loss was easy to adapt for dynamic data and still achieved impressive results, the optimal parameters for dynamic MRI might not have been found yet.

*Which transformations can we apply in Dynamic MRI?*
The temporal symmetry over the time dimension showed improvements particularly at low noise level. We recommend keeping this transformation. Regarding the proposed group transformation with the different affine transformations that evolves over time, we can't conclude as brain MRI might not be the best scenario to test it, and we didn't implement it.

*Is EI sensitive to a limited number of transformations over the temporal axis (with comparison to a combination of transformations in spatial 3D) ?*
We can only partially conclude. To the question "is it worthy to use only one transformation compared to zero", the answer is yes. The time flip showed a relative gain of knowledge and therefore demonstrated that even with only one transformation, we can gain more information (compared to 0). However, we cannot conclude to the question "would we gain more information with a group of spatial transformations" because of the difficulty to build a full 3 spatial dimensions model due to hardware limitations.

### 9.3.3 Observations

Other phenomenons observed and attempts behaviors must be highlighted:

- The SURE loss becomes negative very quickly. It is very surprising to always obtain a negative SURE. We can't pronounce ourselves if it had a big impact on the reconstruction or not as we were not able to keep it positive. Still, even when the term was inferior to 0, it kept decreasing progressively. Possible reasons to this negative loss are exposed in 10.

- The EI loss term fluctuates a lot from one epoch to another. It is not a good thing and shows a lack of stability. We chose to use "no grad" to True in the EI loss due to the weird results obtained in 2D EI MRI. However, a very late attempt using "no grad" to False at $\sigma = 0.05$ showed way more stability in the EI loss without creating unexpected structures (holes) as in 2D. Then, if the EI loss kept decreasing in both 2D and dynamic models, it didn't hallucinate in dynamic, and therefore it might be interesting to continue the researches in that way.

- The REI model is worse at low noises than simple EI. As the noise is very low, we can think that the REI isn't suited for such a situation. Maybe $\text{EI}_1$ or $\text{EI}_2$ would work better in these conditions. No attempt with $\text{EI}_1$ or $\text{EI}_2$ has been made. However a single attempt using REI at noise level 0.01, but with a $\sigma$ of 0.007 in the SURE loss, showed an increase of 0.5 dB (31.7 dB) compared to $\sigma = 0.01$ but is still far from the vanilla EI result. It finished with divergence.

- Using a batch of one is very noisy and the training is sensitive to the last iteration of an epoch, influencing a lot the final model.

- Similarly to 2D SSLEI models, using only one transform was sufficient to prove EI and REI skills and increasing the number of transformations just increased the computation time.

- Going from 128x128x128 volumes to 256 ones was really hard. Models were way less sensitive at lower resolution, surely due to the averaging pooling that was smoothing the coefficients. These smoothed coefficients are easier to find for the model than non-smoothed ones. **The higher the resolution, the harder for SSLEI models to reconstruct**.

### 9.3.4 Further Work

This research was mostly an experiment to evaluate the potential of SSLEI in dynamic MRI imagery. It is still possible to improve the models by working on different aspects and attempts. A non-exhaustive list is proposed:

- Lot of parameters, lot of work! There is a need of identifying the impact of each parameter over the reconstruction. Simply changing the value of one parameter (no grad, $\tau$ in the SURE loss,

parameters to learn...) has lot of impact on the model. As unrolled models initially aim to solve this aspect (fixing parameters) and pretend to learn the whole denoiser with deep denoising, it is a limitation at the moment. We need to understand how a parameter impacts the model.

- We must identify the issue with the negative loss. It is a big issue regarding the very concept itself. An option might be to clip the gradient in the SURE loss, to avoid strong divergence. An attempt showed similar results without resulting in a negative loss.
- In REI, the PSNR keeps increasing as long as the EI loss is also decreasing. If the total loss decreases because of the SURE loss, then the PSNR might decreases (and quite suddenly).
- This experimentation was based on a single denoiser architecture which might not be optimal. U-Net being known to be a good denoiser for imagery (and often used), developing a 3D U-Net is very next step.
- Using smooth dynamical transforms movements was proposed in 6.2. Finding and trying new transformations is still of main interest. The proposed group action with a very smooth movement between two close points in the affine group presented in 6.2 can be tested with accurate interpolation in the movement. It is not as intuitive as a simple helical movement over time (linear rotation). In scenarios with moving structure such as cardiac MRI, and not rigid (as the brain in MRI) it can have lot of sense.
- We insist on the need to assess the impact of having only one transformation compared to more over the time axis. More resources is desirable to learn unrolled SSLEI models in 3 spatial dimensions (and not 2D+t), using combination of rotations, flips etc. over the 3 spatial axis. As patching is not possible in k-space, we are limited. First attempts could be done using 128x128x128 volumes instead of whole resolution.
- We emulated time with a movement over the plan transaxial axis. In fMRI, we only have a fixed image (or volume) with time-varying intensities and not time-varying structures! It might be easier for a model to learn the reconstruction of time-varying intensities rather than structures. Other denoiser might be preferred then, we didn't look for these information yet.
- We used exactly 20 patches of depth 8 to reconstruct the initial depth of 160 slices. It might be interesting to use more patches that overlap each other especially to help having smoother transition from one small patch to another. This suggests we can use the same patches but adding ones from slices 4 to 12 for example to gain smoothing information at the transition between slice 8 and 9. Then using half of the mean of the superposition patches and the initial patches might result in a better reconstruction.

# Chapter 10

# Discussions

## 10.1 Peak Signal-to-Noise Ratio

### 10.1.1 Which PSNR ?

In Magnetic Resonance Imaging real physics, since the signal induced in the coil has a phase, as well as a magnitude, the values in k-space are complex. This is why we use 2 channels in MRI. However, in our situation, we only have access to the magnitude of the volume, as we work with the QTIM dataset, which is composed of NifTi files containing only the voxel intensities. Consequently, we do not have the imaginary part of the signal.

To remain consistent with the real physics of MRI, we still employ two-channel reconstruction models. In this setup, the first channel represents the real part (the magnitude), while the second channel, which should ideally contain the imaginary part (phase), actually only contains noise (due to Deep Inv MRI physics) but without any meaningful data. This raises the question of whether we should use the traditional PSNR or a complex PSNR for evaluating the reconstructions.

PSNR is typically used for evaluating the quality of reconstructions by comparing the original and reconstructed images in terms of their pixel intensity values. Given that the imaginary part does not contain meaningful data and only comprises noise, using a standard PSNR on the magnitude images might be more appropriate. This approach focuses on the actual data of interest (the magnitude) and ignores the noise present in the imaginary channel. However, if the goal is to consider the noise characteristics and their impact on the overall reconstruction, a complex PSNR should be considered, taking into account both channels.

The Deep Inv library containing already

### 10.1.2 Relevance of the metric in neuroimaging

During the implementation, we previously mentioned the bias of the large black areas around the brain and the complexity of fine structures within the brain. These factors can significantly impact the calculated PSNR and may not accurately reflect the true quality of the image reconstruction.

In brain MRI images, there are often large regions outside the brain that appear completely black due to the absence of structures. These black areas have zero or near-zero intensity values even with noise. Since PSNR is based on the mean squared error (MSE) between the original and reconstructed images, the presence of large uniform areas with zero difference can artificially inflate the PSNR value, giving a misleading impression of high reconstruction quality. This is because the MSE is averaged over all pixels, and the zero-difference regions (black areas) reduce the overall error, thus increasing the PSNR.

Also, brain MRI images contain intricate complex and fine structures. These are those structures we really want to reconstruct. As a result, even if the reconstruction is poor in these critical areas, the

impact on the overall PSNR might be minimal due to the dominance of the black regions in the MSE calculation. This means that a high PSNR value might not necessarily correspond to a high-quality reconstruction of the important anatomical details in the brain.

## 10.2 DnCNN Batch Norm

In the original paper [2], the architecture of a DnCNN typically includes batch normalization layers (and initialisation). During the experimentation, tests have been made using this method which aims to stabilize and to fasten the training of the model. It indeed showed way better results over first epochs, adding only small computation time. Instead of a PSNR of $\approx$ 26.5 dB (with $\sigma = 0.05$) on first epoch with EI loss, the model already had $\approx$ 30.5 dB using batch norm. However, this fast convergence led to fast overfitting. After 10 epochs only, if the training reached high PSNR values, the test PSNR was still several dB lower. Maybe this difference would be smaller after 100 epochs but we didn't led the test until this point[1]. Also, adding batch normalization layers increases the memory usage and can saturate the GPU.

## 10.3 SURE Loss

We exposed in 9.3 the issues with the negative SURE Gaussian loss in 3D SSLEI models. As the SURE loss is made of two terms, the MC (measurement consistency) from one side and a (supposedly unbiased) Monte-Carlo estimator of the divergence of the projected reconstruction operator (see REI: 5.3.2). Since we target an estimator of MSE (which is positive), we were expecting a not systematically negative value. How to explain this result ? Multiple factors seem able to contribute.
One can be the lack of precision of 32 bits tensors in 3D. As we use two tensors with very small difference in the Monte-Carlo approximation of the divergence, the computation of their difference can face precision issues and 64 bits tensors might bring better results. This can be significant if some values of random finite differences become large.
The parameter $\tau$ should also be tune with precaution, but we did not notice any different behaviour of SURE loss by changing its value from 0.001 to 0.1 (we kept 0.01). In addition only one Monte-Carlo sample is used for the expectation, considering that terms $h_\theta(y + \tau b)$ are "approximately independent" [39] (page 1545). But is this assumption still available for MSE estimation of the projected operator in k-space ?
Furthermore, theory of SURE estimate of MSE relies on the fact that all derivatives of the projected reconstruction operator remain bounded (theorem 1 in [6]). But can we ensure that these derivatives do not become very large during the optimization process ?
Finally, we used batches of size 1 only, which can lead to very noisy gradients for the SURE loss. We can want to increase the batch size for more stability and less noise in the loss computation... but then the GPU limitations cause issues again. Also we can increase the number of random drawings of the Monte-Carlo part but this considerably increase the computation time.

## 10.4 Rotations and flips in NeuroImaging

The original research suggested to use rotations and flips over the spatial dimensions. This is what was previously used in MRI experiments in [4, 1] for knee MRI. A knee MRI image, while being somewhat symmetrical in terms of general shape, shows significant asymmetry due to the unique structures within each knee. In this case, flips and rotations generate transformations adding extra information. However, the human brain is bilaterally symmetrical, with the left and right hemispheres being mirror images of each other to a large extent. Most of the brain structures exhibit a high degree of symmetry across the midline (the longitudinal fissure), when seen from above (axis we chose for the study). Flipping the image across the midline will not introduce new information but rather mirror the existing information.

[1] This issue was raised after catching an issue in the original Deep Inv library "Trainer" class, which was always computing the metrics in a training mode and not an evaluation mode.

This is why it was preferred to use the newly introduced affine transformations group that incorporates a combination of rotation, flip, shearing and scaling all in one (also shift but it is not necessary to use it).

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